

Decomposing the Bloch–Kato Fundamental Line: Adequacy and No-Go Results for the BSD Invariants

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Abstract

The Birch–Swinnerton-Dyer formula for an elliptic curve $E/\mathbb{Q}$ is a product of pieces of different kinds: a period, a regulator, Tamagawa numbers, torsion and the order of the Tate–Shafarevich group, which in the Bloch–Kato formulation are joined with local p -adic data and orientation conventions in one determinant line. Each piece is the image of richer data. We separate these data into five columns (height/regulator, analytic/period, finite source, local p -adic, and determinant assembly) and ask, column by column, which data a given number remembers. The test is a finite Schur complement, which vanishes exactly when the desired readout is a linear function of the admitted data. The full data of each of the first four columns are adequate for their readouts, and the five columns together are adequate for an orientation-normalized Bloch–Kato scalar, which forgets the orientation. Along the way we fix three normalization points: the invariant finite factor is the order of a paired group, not a numerical Pfaffian of its Cassels–Tate pairing; on the finite local condition the relevant map is the logarithm, not the dual exponential; and the assembled value must not be reused as a factor. We then identify readouts that lose information. The most concrete is arithmetic: the curves 571a1 and 2045b1 have the same rank, Tamagawa numbers, torsion and $\dim \text{Sha}[2]$, but 2-primary Tate–Shafarevich groups of orders 4 and 16. Other results, proved on explicit models, concern the scalar finite factor, Gram matrices in rank at least two, and finite lists of curves and of primes. We also identify the normalization $\kappa_r = \# \text{Sha} / \# E(\mathbb{Q})_{\text{tors}}^2$ under which a cascade form of the leading term agrees with the standard one. The comparisons from the columns to the Bloch–Kato line are stated as external inputs. Nothing here proves BSD. Many statements are checked in Lean 4, and each is labelled by the exact extent of that check.

Keywords. Birch–Swinnerton-Dyer conjecture; Bloch–Kato fundamental line; elliptic curves; Tate–Shafarevich group; Cassels–Tate pairing; Schur complement; regulator; formal verification (Lean); Six Birds Theory; emergence calculus.

Conventions. $E/\mathbb{Q}$ is an elliptic curve, $M_E = h^1(E)(1)$, $r = \text{rank } E(\mathbb{Q})$ and $r_{\text{an}} = \text{ord}_{s=1} L(E, s)$. Curve labels are Cremona labels.

1 Introduction

For an elliptic curve $E/\mathbb{Q}$ of analytic rank r_{an} , the Strong Birch–Swinnerton-Dyer formula [3, 30] compares two numbers:

$$\frac{L^{(r_{\text{an}})}(E, 1)}{r_{\text{an}}!} \quad \text{and} \quad \Omega_E \text{Reg}(E/\mathbb{Q}) \frac{\# \text{Sha}(E/\mathbb{Q}) \text{Tam}(E)}{\# E(\mathbb{Q})_{\text{tors}}^2}.$$

The right-hand side is visibly a product of pieces of different kinds: a period, a regulator computed from heights of rational points, Tamagawa numbers, torsion, and the order of the Tate–Shafarevich

group. In the Bloch–Kato formulation [5, 13] these pieces are assembled into a single determinant line, the fundamental line, together with local p -adic data and sign and orientation conventions.

This paper does not ask whether the formula is true. It asks a more modest question about its ingredients: each factor is the image of richer data (a lattice with a pairing, a finite group with a pairing, a local Galois representation), and one may ask which of those data a given number still remembers. We separate the ingredients into five *columns*:

1. height/regulator: the Mordell–Weil lattice and its Néron–Tate pairing;
2. analytic/period: the leading coefficient of $L(E, s)$ at $s = 1$ and the real period;
3. finite source: Tamagawa numbers, torsion, the p -primary parts of Sha and the Cassels–Tate pairing;
4. local p -adic: data at a prime p , of two kinds treated here, ordinary data at a prime of good ordinary reduction and signed data at an odd prime of good supersingular reduction with $a_p = 0$;
5. determinant assembly: the orientation and sign conventions used to multiply the other four in one line.

For each column we test whether its native data are enough to recover the intended readout, and we identify readouts that are not.

The adequacy residual. All tests in the paper use one finite-dimensional device. Let $x \in \mathbb{R}^n$ be coordinates for the data of a column, let $D : \mathbb{R}^n \rightarrow \mathbb{R}^m$ be the *readout* (the quantity one wants), and let $L : \mathbb{R}^n \rightarrow \mathbb{R}^k$ be the *source* (the data one admits). With the standard inner product on $\mathbb{R}^n$ (the matrix $C = I$ that appears in the statements below), form the Gram blocks

$$K_{LL} = LL^\top, \quad K_{DL} = DL^\top, \quad K_{DD} = DD^\top,$$

and the Schur complement [38, 15]

$$\Xi_C(D \mid L) = K_{DD} - K_{DL} K_{LL}^\dagger K_{LD},$$

where $K_{LL}^\dagger$ is the Moore–Penrose inverse. If P_L is the orthogonal projector onto the row space of L , then $\Xi_C(D \mid L) = D(I - P_L)D^\top$, a positive semidefinite matrix. It vanishes exactly when $D(I - P_L) = 0$, that is, when every row of D lies in the row space of L , or equivalently when $D = ML$ for some matrix M . We say that the source is *adequate* for the readout when $\Xi_C(D \mid L) = 0$: the readout is then a linear function of the admitted data. When D is the identity, $\Xi_C(D \mid L) = I - P_L$ and its trace counts the directions the source does not see.

In this language the results are small, explicit computations. For example, the readout of a scalar finite factor is a single row $s = (1, -2, 1, 0)$ in four finite coordinates, and $\text{tr}(I_4 - P_s) = 4 - 1 = 3$; at a supersingular prime, a source that keeps the signed comparison coordinate but replaces the two signed coordinates by their sum leaves $\text{tr}(I_3 - P_{\text{uns}}) = 3 - 2 = 1$. The calculations are elementary by design. Their purpose is bookkeeping: they record which coordinates must be present before a datum can be used as a source for a given readout.

An example. For the curve 11a1 [10, 31],

$$\frac{L(E, 1)}{\Omega_E} = 0.2 \quad \text{and} \quad \frac{c_{11} \# \text{Sha}(E/\mathbb{Q})}{\#E(\mathbb{Q})_{\text{tors}}^2} = \frac{5 \cdot 1}{5^2} = 0.2.$$

As a scalar check this is one equality. In the five columns it is several facts: the height column is empty (rank 0); the finite column records that the value $5/25$ comes from a Tamagawa number and from torsion, not from Sha; and the local columns record, prime by prime, which local theory applies. The scalar equality is compatible with, but coarser than, these facts. A valuation can sit in the Tamagawa factor or in Sha and contribute the same exponent to the scalar; two signed local directions can have the same unsigned sum; a Gram matrix computes the regulator but says nothing about where the points come from.

Results. Section 2 sets up the five columns, the assembly into the Bloch–Kato line, and an additive splitting of the comparison defect. Sections 3 to 7 treat the columns one at a time. In the first four, the full native data are adequate for the column readout; in the assembly column, the four arithmetic columns fall short of the assembled target by exactly the orientation direction. Each column section ends with the comparison to the Bloch–Kato line that the column requires as external input. Along the way we fix three normalization points: the order of a paired finite group, not a numerical Pfaffian of its pairing, is the invariant finite factor; on the finite local condition the relevant map is the logarithm, not the dual exponential; and the orientation enters the assembly as an independent input, so that the assembled value must not be reused as a factor. Section 8 contracts the five columns to an orientation-normalized scalar readout and shows that the typed data are adequate for it, while this readout forgets the orientation coordinate; identifying it with the arithmetic Bloch–Kato scalar requires the five transports and, in addition, a convention making that scalar independent of the orientation.

Sections 5 and 9 to 11 contain the “does not determine” results (Table 2). The most concrete is arithmetic: the curves $571a1$ and $2045b1$ have the same rank, Tamagawa numbers, torsion and $\dim \text{Sha}[2] = 2$, but 2-primary Tate–Shafarevich groups of orders 4 and 16 (Example 5.5). The others are formal and are proved on explicit models: the scalar finite factor forgets which block carries a valuation; the Gram matrix forgets the provenance of a basis; and finite lists of curves or of primes do not decide statements about all curves or all primes. Section 12 introduces the exterior algebra vsr_r as the formal shape of a rank- r source, and Section 13 identifies the constant $\kappa_r = \# \text{Sha} / \# E(\mathbb{Q})_{\text{tors}}^2$ that makes a cascade form of the leading term agree with the standard one. Section 14 lists what the paper does not claim.

Context. The vocabulary of columns, readouts and residuals is a specialization of the adequacy calculus of the Six Birds framework [36, 34, 35]. No knowledge of that framework is assumed; the definitions above are all that is used. The companion paper [33] uses the present apparatus, in particular the normalization of κ_r , in a conditional treatment of the scalar BSD identity. Many statements are checked in Lean 4 without external mathematical libraries; the footnotes say, statement by statement, whether Lean proves the statement as written, a narrower model version, or only records it as a conditional schema, and Appendix A gives the details.

2 Decomposition: the typed five columns

The introduction separated the BSD product into five kinds of data. We now make the separation precise. The aim is not to replace the Bloch–Kato determinant line by five unrelated objects. It is to keep track of five native sources *before* they are multiplied together, so that each can be tested on its own.

Fix an elliptic curve $E/\mathbb{Q}$ and put $M_E = h^1(E)(1)$. The five columns are:

1. the *height/regulator* column: the Mordell–Weil lattice and its Néron–Tate pairing;

Table 1. The five typed columns and their adequacy residuals. Each row separates the native source data from the determinant-line target before the scalar BSD product is read out.

Column	Carrier map	Residual	Adequate when
Height/regulator	$\rho_{\text{ht/reg}}^{\mathbb{R}}$ from the Mordell–Weil lattice and Neron–Tate pairing	$\Xi_C(D \mid L)$ for $d(\log \det)_H(\delta H) = \text{tr}(H^{-1}\delta H)$	the full height pairing is admitted, so the identity source kills the Schur residual.
Analytic/period	$\rho_{\text{an/period}}^{\mathbb{R}}$ from (A_E, Ω_E^+)	$\Xi_C(D \mid L)$ for $\log A_E - \log \Omega_E^+$ ($A_E \neq 0$)	both logarithmic coordinates are present; either coordinate alone leaves a one-dimensional residual.
Finite/Tamagawa	$\rho_{\text{finite}}^{\mathbb{R}}$ from Tamagawa, torsion, $\text{Sha}[p^\infty]$, and Cassels–Tate data	$\Xi_C(D \mid L)$ for the typed finite-source readout	the full finite-source tower, including the Cassels–Tate pairing, is retained.
p -adic	ρ_p from ordinary or signed local data	$\Xi_C(D \mid L)$ for the ordinary/signed local readout	the appropriate ordinary or signed local source is admitted, not merely an unsigned public shadow.
Determinant assembly	ρ_{det} for orientation, sign, and unit compatibility	$\Xi_C(D \mid L)$ for the assembled determinant comparison	the determinant-orientation coordinate is included with the four arithmetic columns.

2. the *analytic/period* column: the leading Taylor coefficient of $L(E, s)$ at $s = 1$ and the real period;
3. the *finite-source* column: Tamagawa numbers, rational torsion, the p -primary parts of Sha, and the Cassels–Tate pairing;
4. for each prime p , the *local p -adic* column: ordinary or signed local data at p , in the cases treated in Section 6;
5. the *determinant-assembly* column: the orientation, sign and unit conventions needed to place the other four columns in one determinant line.

Table 1 and Figure 1 summarize the columns and the common adequacy test applied to each of them.

The Bloch–Kato target is the determinant-line package attached to M_E [11, 5, 13]: the motive, its fundamental line, the zeta element, and a chosen trivialization. The source object is the *integrated BSD carrier*, which keeps the five kinds of data in separate slots. The first structural map assembles the source into the target.

Definition 2.1 (Bloch–Kato component-assembly map). For $M_E = h^1(E)(1)$, let

$$C_{\text{BSD}}(E) = (C_{\text{ht}}(E), C_{\text{an}}(E), C_{\text{fin}}(E), \{C_p(E)\}_p, C_{\text{det}}(E))$$

be the integrated BSD carrier, with height/regulator, analytic/period, finite-source, local p -adic and determinant-assembly slots. The source-side slot $C_{\text{det}}(E)$ holds an orientation datum and is distinct from the Bloch–Kato target

$$C_{\text{BK}}(E) = (M_E, \Delta_f(M_E), z_{\text{BK}}(M_E), \tau_{\text{BK}}(M_E)).$$

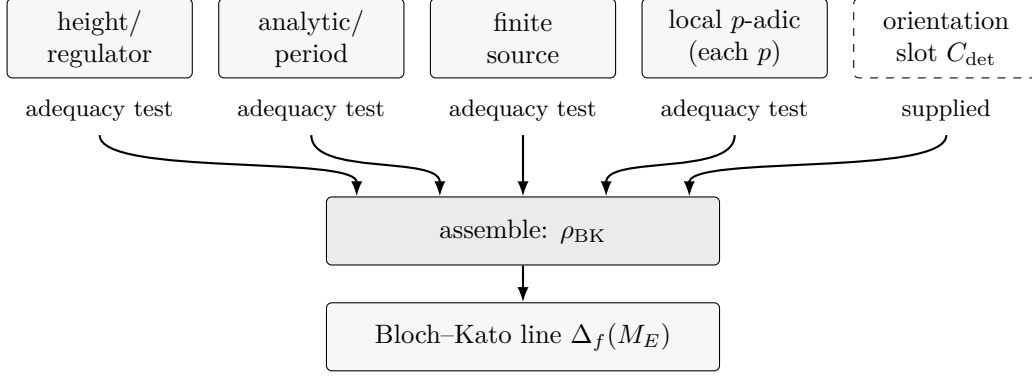


Figure 1. The five slots of the integrated BSD carrier. Each of the four arithmetic columns is tested for adequacy on its own, by the Schur residual $\Xi_C(D \mid L)$ of Section 1. The assembly then combines the four column values with an independent orientation datum; the assembled value is not itself fed back in as a fifth factor (Section 8).

Suppose given column maps ρ_{ht} , ρ_{an} , ρ_{fin} , ρ_p and ρ_{det} into the target, and an assembly operation that combines one value from each column. The component-assembly map

$$\rho_{\text{BK}} = \text{assemble}(\rho_{\text{ht}}, \rho_{\text{an}}, \rho_{\text{fin}}, (\rho_p)_p, \rho_{\text{det}}) : C_{\text{BSD}}(E) \longrightarrow C_{\text{BK}}(E)$$

applies each column map to its own slot and then assembles the results. Informally, ρ_{ht} sends height and regulator data to the global cohomological side; ρ_{an} sends the leading term and period to the analytic (zeta-element) side; ρ_{fin} sends Tamagawa, torsion, $\text{Sha}[p^\infty]$ and Cassels–Tate data to the finite determinant side; ρ_p sends local p -adic data to the p -primary side; and ρ_{det} supplies the orientation in the common line.

The definition organizes the source; it constructs neither the arithmetic column maps nor the assembly operation. Those are exactly the five comparison transports recorded as obligations in the column sections below (Obligations 3.3 to 6.4 and 5.6). Once a common trivialization is chosen, the total comparison defect can be split into one contribution per column.

Definition 2.2 (Bridge-defect equation). Choose a trivialization of the determinant-line target in which the component comparisons take values in a common additive group, and write $\Delta_{\text{BSD}}^{\text{BK}}$ for the total defect of the BSD comparison. The bridge-defect equation is the decomposition

$$\Delta_{\text{BSD}}^{\text{BK}} = E_{\text{ht/reg}} + E_{\text{an/period}} + E_{\text{finite}} + \sum_p E_p + E_{\text{det}}$$

into the height/regulator, analytic/period, finite-source, local and determinant-assembly defects. The local sum is required to be finite: only finitely many E_p are nonzero, and the sum of the local defects is zero when every E_p is zero. Passing between this additive form and the multiplicative form (a total ratio equal to the product of five component ratios) requires a logarithm or trivialization convention that also accounts for signs and units; we do not fix one here.

Theorem 2.3 (Component-killing). *Assume the bridge-defect equation holds in the chosen trivialization. If*

$$E_{\text{ht/reg}} = 0, \quad E_{\text{an/period}} = 0, \quad E_{\text{finite}} = 0, \quad E_p = 0 \text{ for every } p, \quad E_{\text{det}} = 0,$$

then $\Delta_{\text{BSD}}^{\text{BK}} = 0$. Equivalently, if $\Delta_{\text{BSD}}^{\text{BK}} \neq 0$, then either some component defect is nonzero or the componentwise comparison (the bridge-defect equation together with its finiteness condition) is not available.¹

Proof. By the finiteness condition, $\sum_p E_p = 0$ when every E_p vanishes. Substituting the five vanishing defects into the bridge-defect equation gives $\Delta_{\text{BSD}}^{\text{BK}} = 0$. The second statement is the contrapositive. $\square$

The next result is the first of several statements in this paper of the form “a public number does not determine the typed datum behind it”. It concerns the scalar finite factor of the BSD formula.

Theorem 2.4 (The finite scalar factor forgets where a valuation sits). *Fix a prime p . A finite-source record at p is a triple of integers (s, t, u) , standing for $(v_p(\# \text{Sha}), v_p(\prod_{\ell} c_{\ell}), v_p(\#E(\mathbb{Q})_{\text{tors}}))$, with s even, together with a Cassels–Tate datum θ from some set with at least two elements. The valuation of the scalar finite factor*

$$\phi_{\text{fin}} = \frac{\# \text{Sha}(E/\mathbb{Q}) \prod_{\ell} c_{\ell}}{\#E(\mathbb{Q})_{\text{tors}}^2}$$

defines the readout $q_p(s, t, u, \theta) = s + t - 2u$ on records. This readout is not injective: distinct records with different nonzero blocks have the same value, and q_p does not depend on θ at all.²

Proof. The parity condition on s reflects the fact that, when $\text{Sha}(E/\mathbb{Q})$ is finite, its order is a square, because the Cassels–Tate pairing is alternating and nondegenerate [8, 27]. The records $(2, 0, 0, \theta)$ and $(0, 2, 0, \theta)$ are distinct: the first places the valuation in the Sha block, the second in the Tamagawa block. Both give $q_p = 2$. Finally q_p depends only on (s, t, u) , so changing θ does not change it. $\square$

For a curve with finite Sha, the formula for q_p is the p -adic valuation of the arithmetic factor ϕ_{fin} . The noninjectivity proved above concerns formal records; we do not assert here that its two witnesses are realized by curves.

The theorem concerns a specific readout. It does not say that no richer arithmetic theorem, given additional source data, can recover the finite datum. This qualification applies to every “does not determine” result in the paper, and we will not repeat it each time.

Nonclaim 2.5 (Decomposition scope). This section does not prove BSD, the Bloch–Kato conjecture, or the equivariant Tamagawa number conjecture [7]. It defines the five-column assembly, proves the component-killing consequence of the bridge-defect equation, and shows that the scalar finite factor forgets information. The comparison transports from the five native columns to the determinant-line target are external arithmetic inputs, recorded separately as obligations; they are not derived here.

3 Height/regulator column

The height/regulator column keeps the Mordell–Weil lattice and its Néron–Tate pairing [25, 29] before any comparison with the global determinant line. Its readout is the regulator, viewed as the coefficient of a determinant.

¹Lean proves the integer-valued implication, with a supplied value for the local sum and the explicit hypothesis that this value is zero when every E_p is zero; finiteness of the support is one way to guarantee that hypothesis. Declaration names for all formal statements are collected in Appendix A.

²Lean proves the statement for this model of the tower, with a two-valued Cassels–Tate tag. Whether the two records below are realized by elliptic curves is not addressed; Section 5 gives an arithmetic example for a different readout.

Definition 3.1 (Real height-regulator map). Let

$$\Lambda_E = E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}}, \quad V_E = \Lambda_E \otimes_{\mathbb{Z}} \mathbb{R},$$

and let $\langle \cdot, \cdot \rangle_{\text{NT}}$ be the Néron–Tate pairing on V_E , with associated map

$$h_{\text{NT}} : V_E \longrightarrow V_E^*, \quad P \longmapsto \langle P, \cdot \rangle_{\text{NT}}.$$

The real height-regulator map sends the triple $(V_E, \Lambda_E, \langle \cdot, \cdot \rangle_{\text{NT}})$ to

$$\rho_{\text{ht}/\text{reg}}^{\mathbb{R}} = \det(h_{\text{NT}}) \in \det(V_E^*) \otimes \det(V_E)^*.$$

If $(P_1, \dots, P_r)$ is a $\mathbb{Z}$ -basis of Λ_E and $H_E = (\langle P_i, P_j \rangle_{\text{NT}})_{1 \leq i, j \leq r}$ is the Gram matrix, the coefficient of this element in the induced basis is $\det H_E = \text{Reg}(E/\mathbb{Q})$, with the empty determinant equal to 1 when $r = 0$. The coefficient does not depend on the basis: a change of basis by $A \in \text{GL}_r(\mathbb{Z})$ replaces H_E by $A^T H_E A$, and $\det(A^T H_E A) = \det(A)^2 \det H_E = \det H_E$ because $\det A = \pm 1$. (A change of basis of finite index $m > 1$ multiplies the coefficient by m^2 , so integrality of the basis matters.)

The adequacy test for this column asks whether the full height matrix is enough to recover the first-order variation of $\log \text{Reg}$. It is, for a simple reason: the readout is a linear function of the variation, and the full source sees every coordinate of the variation.

Theorem 3.2 (Height tangent Schur collapse). *Let $r \geq 1$ and let H be a positive definite real symmetric $r \times r$ matrix. Identify a symmetric variation δH with its coordinate vector $x = \text{vech}(\delta H) \in \mathbb{R}^n$, $n = r(r+1)/2$, listing the entries δh_{ij} with $i \leq j$. Then*

$$D_H(\delta H) = d(\log \det)_H(\delta H) = \text{tr}(H^{-1} \delta H) = d_H x,$$

where the row d_H has entry $(H^{-1})_{ii}$ at each diagonal coordinate and $2(H^{-1})_{ij}$ at each off-diagonal coordinate $i < j$. Take the audit energy $C = I_n$ on these coordinates and the full source $L_{\text{full}}(x) = x$. Then

$$K_{DD} = \|d_H\|^2, \quad \Xi_C(D \mid L) = d_H d_H^T - d_H I_n d_H^T = 0.$$

For $r = 0$ the tangent space is zero, $n = 0$, and $K_{DD} = 0$: the rank-zero row is vacuous.³

Proof. Since H is invertible, $\det(H + t \delta H) = \det H \cdot \det(I + tA)$ with $A = H^{-1} \delta H$. In the permutation expansion of $\det(I + tA)$, the identity permutation contributes $1 + t \text{tr} A + O(t^2)$, and every other permutation moves at least two indices and contributes $O(t^2)$. Since $\det H > 0$, the chain rule gives $d(\log \det)_H(\delta H) = \text{tr}(H^{-1} \delta H)$. Writing $G = H^{-1}$, which is symmetric,

$$\text{tr}(G \delta H) = \sum_{i,j} G_{ij} \delta h_{ji} = \sum_i G_{ii} \delta h_{ii} + \sum_{i < j} 2G_{ij} \delta h_{ij},$$

because the off-diagonal coordinate δh_{ij} occurs twice in the symmetric matrix δH . This is $d_H x$.

For the Schur residual, the source block is $K_{LL} = I_n$, whose Moore–Penrose inverse is I_n , and the cross blocks are $K_{DL} = d_H$ and $K_{LD} = d_H^T$. Hence $K_{DL} K_{LL}^\dagger K_{LD} = d_H I_n d_H^T = d_H d_H^T = K_{DD}$, and $\Xi_C(D \mid L) = 0$. If $r = 0$, the row d_H is empty and the empty sum $d_H d_H^T$ is 0. $\square$

³Lean proves the algebraic core: for every row d of every length over any commutative ring, the identity source gives Schur residual zero, and the identity matrix is the unique matrix satisfying the four Penrose equations at the identity. For $r = 2$ it also proves that the displayed row computes $\text{tr}(H^{-1} \delta H)$ when $\det H \neq 0$. The calculus step identifying the row with the derivative of $\log \det$, and the general-rank row, are proved here on paper.

For example, when $r = 2$ and $H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$ with $\Delta = ac - b^2 > 0$, the row in the coordinates $(\delta h_{11}, \delta h_{12}, \delta h_{22})$ is

$$d_H = \left(\frac{c}{\Delta}, -\frac{2b}{\Delta}, \frac{a}{\Delta} \right),$$

as one also sees from $\det(H + t\delta H) = \Delta + t(c\delta h_{11} - 2b\delta h_{12} + a\delta h_{22}) + O(t^2)$. When $r = 1$, $d_H = 1/\text{Reg}(E/\mathbb{Q})$ and $K_{DD} = \text{Reg}(E/\mathbb{Q})^{-2}$; for the curve 37a1, the regulator 0.0511114... recorded in the LMFDB [31] gives $K_{DD} = 382.79\dots$. The choice $C = I_n$ refers to the chosen vech coordinates; it is not invariant under changes of lattice basis, and only the vanishing of $\Xi_C(D \mid L)$, not the value of K_{DD} , is used later.

Obligation 3.3 (Height/regulator transport). The Bloch–Kato height/regulator component requires a comparison

$$\beta_{\text{BK,ht}} : \det(V_E^*) \otimes \det(V_E)^* \longrightarrow \Delta_f(M_E)_{\text{ht/reg}}$$

identifying the Néron–Tate determinant with the Beilinson–Deligne regulator normalization [1]. It must fix the coefficient and lattice conventions, the archimedean Deligne-cohomology normalization, and the orientation and trivialization, and it must be compatible with the finite and determinant-assembly columns without absorbing them. This comparison is external arithmetic input. It is not derived from the finite calculation above, which only shows that the full height matrix is adequate for the first-order regulator readout.

Warning 3.4 (Rank-zero carriers). For a curve of rank 0 the height/regulator column is empty: there is no nonzero height variation, and the regulator is the empty determinant 1. The curve 571a1, which reappears in Section 5 because its Tate–Shafarevich group has nontrivial 2-primary part, is such a curve: a certified 2-descent bounds its rank above and below by 0 (details in Section 5). It should not be confused with the rank-two curve 571b1 of the same conductor. Rank-two height/regulator examples in this paper use 389a1.

4 Analytic/period column

The analytic/period column keeps the analytic leading coefficient and the real period apart from the height, finite and local data. Its native data form a pair of numbers, and its readout is their quotient.

Definition 4.1 (Real analytic/period map). By modularity [37, 6], $L(E, s) = L(f_E, s)$ extends to an entire function. Let $r_{\text{an}} = \text{ord}_{s=1} L(E, s)$ and

$$A_E = \frac{L^{(r_{\text{an}})}(E, 1)}{r_{\text{an}}!} \neq 0$$

be the analytic leading coefficient, that is, the leading Taylor coefficient of $L(E, s)$ at $s = 1$. Let $\Omega_E^+ > 0$ be the real period in the BSD normalization: the integral of $|\omega_E|$ over $E(\mathbb{R})$ for a Néron differential ω_E . The native analytic/period family is the pair

$$L_{\text{an/per}}(E) = (A_E, \Omega_E^+),$$

and the real analytic/period map is

$$\rho_{\text{an/period}}^{\mathbb{R}} : (A_E, \Omega_E^+) \longmapsto (A_E/\Omega_E^+) e_{\text{an/per}},$$

where $e_{\text{an/per}}$ is a basis vector of a real line. The column contains no height, torsion, Tamagawa or Sha data.

The adequacy test asks whether the pair is enough to recover the first-order variation of the quotient. In logarithmic coordinates the absolute value of the quotient becomes a difference; its sign is not a tangent direction and is recorded separately, as part of the orientation data of Section 7.

Theorem 4.2 (Analytic/period tangent Schur collapse). *In the chart*

$$x = (x_A, x_\Omega) = (\log |A_E|, \log \Omega_E^+) \in \mathbb{R}^2,$$

take $C = I_2$, the full native source $L_{\text{full}}(x) = x$, and the readout $D(x) = x_A - x_\Omega = \log |A_E/\Omega_E^+| = dx$ with $d = (1, -1)$. Then $K_{LL} = I_2$, $K_{DD} = dd^\top = 2$ and

$$\Xi_C(D \mid L) = 2 - d I_2 d^\top = 0.$$

With no source admitted the residual is 2; with only x_A admitted it is 1; with both coordinates admitted it is 0.⁴

Proof. The readout energy is $K_{DD} = 1^2 + (-1)^2 = 2$. With the full source, $K_{LL} = K_{LL}^\dagger = I_2$, $K_{DL} = d$ and $K_{LD} = d^\top$, so the projected energy is $d I_2 d^\top = 2$ and $\Xi_C(D \mid L) = 0$. With no source nothing is projected away and the residual is 2. If only x_A is admitted, d splits orthogonally as the seen part $(1, 0)$ plus the unseen part $(0, -1)$, and the residual is the squared length 1 of the unseen part. $\square$

The theorem is elementary, and that is the point: a single observed coordinate cannot recover a quotient of two independent quantities, while the pair can. The arithmetic content of the column lies entirely in the comparison below.

Obligation 4.3 (Analytic/period transport). The Bloch–Kato analytic component requires a comparison

$$\beta_{\text{BK,an}} : L_{\text{an/per}}(E)_{\mathbb{R}} \longrightarrow \Delta_f(M_E)_{\text{an/per}}$$

identifying A_E/Ω_E^+ with the zeta-element factor of the fundamental line. It must fix the rational and integral normalizations, the Betti–de Rham comparison and the real orientation, the leading-term convention in positive rank, and the separation from the other four columns. We treat this comparison as external input in every rank. In rank 0 the analytic quotient $L(E, 1)/\Omega_E^+$ is a classical object, but matching it to the zeta-element factor in the conventions above is part of the comparison, not a consequence of the Schur calculation.

5 Finite-source column

The finite-source column keeps the finite arithmetic data before they are contracted to the scalar finite factor $\# \text{Sha} \prod_v c_v / \# E(\mathbb{Q})_{\text{tors}}^2$. Its tower separates Tamagawa numbers [28], rational torsion, the p -primary parts of Sha, and the Cassels–Tate pairing [8, 27]. The pairing is part of the source: it determines the structure of each p -primary group, while the scalar factor only sees orders.

Definition 5.1 (Real finite-source map). The finite-source tower is $C_{\text{fin}}(E) = C_{\text{Tam}} \oplus C_{\text{tors}} \oplus C_{\text{Sha}[p^\infty]} \oplus C_{\text{CT}}$, with native family

$$L_{\text{finite}}(E) = \left((c_v(E))_{v|N}, E(\mathbb{Q})_{\text{tors}}, (S_p, \langle \cdot, \cdot \rangle_{\text{CT},p})_p \right), \quad S_p = \text{Sha}(E/\mathbb{Q})[p^\infty] / \text{Sha}(E/\mathbb{Q})[p^\infty]_{\text{div}},$$

⁴Lean proves these three integer residual computations for the row $(1, -1)$.

where $\langle \cdot, \cdot \rangle_{\text{CT},p} : S_p \times S_p \rightarrow \mathbb{Q}/\mathbb{Z}$ is the perfect alternating Cassels–Tate pairing. Throughout this section we assume that $\text{Sha}(E/\mathbb{Q})$ is finite. Then each divisible part is zero, $S_p = \text{Sha}(E/\mathbb{Q})[p^\infty]$, only finitely many S_p are nontrivial, and $\prod_p \#S_p = \#\text{Sha}(E/\mathbb{Q})$. The real finite-source map is the product of three factors:

$$\rho_{\text{finite}}^{\mathbb{R}} = \rho_{\text{Tam}}^{\mathbb{R}} \otimes \rho_{\text{tors}}^{\mathbb{R}} \otimes \bigotimes_p \rho_{\text{Sha-CT},p}^{\mathbb{R}},$$

with $\rho_{\text{Tam}}^{\mathbb{R}} = \prod_{v|N} c_v$, where $c_v = \#E(\mathbb{Q}_v)/E^0(\mathbb{Q}_v)$; $\rho_{\text{tors}}^{\mathbb{R}} = \#E(\mathbb{Q})_{\text{tors}}^{-2}$; and

$$\rho_{\text{Sha-CT},p}^{\mathbb{R}}(S_p, \langle \cdot, \cdot \rangle_{\text{CT},p}) = \#S_p.$$

The last factor has an integral description. If an alternating integer matrix A presents S_p (that is, $S_p \cong \text{coker } A$) with linking form $(x, y) \mapsto -x^T A^{-1} y \bmod \mathbb{Z}$ equal to the pairing, then $\#S_p = |\det A| = \text{Pf}(A)^2$. For example, the hyperbolic plane $H_n = (\mathbb{Z}/n\mathbb{Z})^2$ with $\langle (a, b), (c, d) \rangle = (ad - bc)/n \bmod \mathbb{Z}$ is presented by $A = \begin{pmatrix} 0 & n \\ -n & 0 \end{pmatrix}$, with $|\text{Pf}(A)| = n$ and $\#H_n = n^2$.

Three numbers attached to H_n should be kept apart: the coefficient $1/n$ of the pairing on a chosen basis, the integral Pfaffian n (up to sign), and the order n^2 . Only the last enters the BSD formula. The first is not an invariant of the paired group: the rational lifts $1/2$ and $3/2$ of the same class in $\mathbb{Q}/\mathbb{Z}$ have different “Pfaffians”, and on H_4 the automorphism $(a, b) \mapsto (3a, b)$ changes the coefficient on the new basis from $1/4$ to $3/4$. The integral presentation removes this ambiguity: a unimodular change of basis B replaces A by $B^T A B$, which changes $\text{Pf}(A)$ by the sign $\det B$ and leaves $\text{Pf}(A)^2$ unchanged.

Theorem 5.2 (Pairing-aware finite Schur collapse). *At a prime p , use the finite tangent coordinates $x = (x_T, x_{\text{tors}}, x_{\text{Sha}}, x_{\text{CT}}) \in \mathbb{R}^4$, where $x_T = v_p(\prod_v c_v)$, $x_{\text{tors}} = v_p(\#E(\mathbb{Q})_{\text{tors}})$, $x_{\text{Sha}} = v_p(\#S_p)$, and x_{CT} is a coordinate for the Cassels–Tate datum. With the full readout $D_{\text{full}}(x) = x$, the full source $L_{\text{full}}(x) = x$ and $C = I_4$,*

$$\Xi_C(D \mid L) = I_4 - I_4 I_4^\dagger I_4 = 0.$$

*Thus the full typed finite tower is adequate for the full finite readout.*⁵

Proof. All four blocks $K_{DD}, K_{LL}, K_{DL}, K_{LD}$ equal I_4 , and the Moore–Penrose inverse of I_4 is I_4 . Hence $K_{DL} K_{LL}^\dagger K_{LD} = I_4$ and $\Xi_C(D \mid L) = I_4 - I_4 = 0$. $\square$

The two results below concern readouts that keep less. The first is the scalar finite factor; the second keeps only the size of the p -torsion of Sha .

Theorem 5.3 (The scalar finite row leaves a three-dimensional residual). *In the coordinates of Theorem 5.2, the valuation of the scalar finite factor is $q = s \cdot x$ with $s = (1, -2, 1, 0)$. The orthogonal projector onto this row is $P_s = s^T (ss^T)^{-1} s$, and the residual against the full readout is*

$$\Xi_{\text{scalar}} = I_4 - P_s, \quad \text{tr}(\Xi_{\text{scalar}}) = 3.$$

*The readout is not injective: the coordinate vectors $(0, 0, 2, h)$, for any h , and $(2, 0, 0, 0)$ are distinct and both give $q = 2$.*⁶

⁵Lean proves the identity calculation on the four declared coordinates. It does not model the finite groups themselves; those enter in Theorem 5.4.

⁶Lean proves the projector trace, the residual trace and the two evaluations over $\mathbb{Q}$. The vectors are formal coordinates; as in Theorem 2.4, arithmetic realization is not asserted.

Proof. Here $ss^\top = 1 + 4 + 1 + 0 = 6$, so $(P_s)_{ij} = s_i s_j / 6$ and $\text{tr } P_s = 6/6 = 1$. Hence $\text{tr}(I_4 - P_s) = 4 - 1 = 3$. The two evaluations are $q(0, 0, 2, h) = 2$ and $q(2, 0, 0, 0) = 2$; the coordinate h does not enter because the last entry of s is 0. $\square$

Theorem 5.4 (Two-torsion size does not determine the paired group).

1. On the two-coordinate model $y = (m, \theta)$, where m records the $\mathbb{F}_p$ -dimension of the p -torsion and θ the remaining structure of the paired group, the dimension-only source $L_{\dim}(y) = (1, 0)y$ leaves the residual $\Xi_{\dim} = I_2 - (1, 0)^\top(1, 0) = \text{diag}(0, 1)$, of trace 1.
2. The groups $H_2 = (\mathbb{Z}/2\mathbb{Z})^2$ and $H_4 = (\mathbb{Z}/4\mathbb{Z})^2$, with the pairings $\langle (a, b), (c, d) \rangle = (ad - bc)/n \bmod \mathbb{Z}$, are finite abelian groups with perfect alternating pairings. Their 2-torsion subgroups both have $\mathbb{F}_2$ -dimension 2, but their orders are 4 and 16. Hence no function of the 2-torsion dimension recovers the order.
3. Conversely, H_4 and $H_2 \oplus H_2$ both have order 16 and carry perfect alternating pairings, but their 2-torsion subgroups have dimensions 2 and 4. Hence no function of the order recovers the 2-torsion dimension.⁷

Proof. For (1), the dimension source keeps the first coordinate and forgets the second, so its projector is $\text{diag}(1, 0)$ and the residual is $\text{diag}(0, 1)$. For (2), H_n is presented by $A = \begin{pmatrix} 0 & n \\ -n & 0 \end{pmatrix}$, and the pairing is perfect because its matrix on the standard basis, $\begin{pmatrix} 0 & 1/n \\ -1/n & 0 \end{pmatrix}$, induces an isomorphism $H_n \rightarrow \text{Hom}(H_n, \frac{1}{n}\mathbb{Z}/\mathbb{Z})$. The 2-torsion of H_2 is H_2 itself, and the 2-torsion of H_4 is $(2\mathbb{Z}/4\mathbb{Z})^2 \cong (\mathbb{Z}/2\mathbb{Z})^2$; both have dimension 2. The orders are 4 and 16. For (3), the orthogonal sum of two perfect alternating pairings is perfect and alternating, $\#(H_2 \oplus H_2) = 16$, and its 2-torsion is the whole group, of dimension 4. $\square$

The theorem is about finite paired groups. The next example shows that the loss it describes occurs for actual Tate–Shafarevich groups, even when several other finite invariants are held fixed.

Example 5.5 (Matched coarse data, different 2-primary Sha). Consider the curves

$$\begin{aligned} 571a1 : \quad y^2 + y &= x^3 - x^2 - 929x - 10595, \\ 2045b1 : \quad y^2 + xy &= x^3 - x^2 - 5470x - 862675. \end{aligned}$$

Their invariants compare as follows.

	571a1	2045b1
rank of $E(\mathbb{Q})$	0	0
Tamagawa numbers	all 1	all 1
$E(\mathbb{Q})_{\text{tors}}$	trivial	trivial
$\dim_{\mathbb{F}_2} \text{Sha}(E/\mathbb{Q})[2]$	2	2
$\text{Sha}(E/\mathbb{Q})[2^\infty]$	$(\mathbb{Z}/2\mathbb{Z})^2$	$(\mathbb{Z}/4\mathbb{Z})^2$
$\# \text{Sha}(E/\mathbb{Q})$	$4u, u \text{ odd}$	$16v, v \text{ odd}$

So every function of the rank, the Tamagawa numbers, the torsion group and $\dim \text{Sha}[2]$ takes the same value on the two curves, while their 2-primary Sha groups, their orders of Sha, and hence their scalar finite factors $\# \text{Sha} / \# E(\mathbb{Q})_{\text{tors}}^2$ differ. The two curves are not matched on everything: their analytic quotients $|L(E, 1)|/\Omega_E^+$ are 4 and 16.

⁷Lean proves the coordinate-model calculation in (1) and the finite-group statement (2), with perfectness checked against every additive character into the cyclic target; (3) is proved in a separate Lean module. The arithmetic example below uses inputs from outside Lean.

The entries of the table are established as follows; we indicate which inputs are external to this paper.

1. *Rank 0 and finiteness of Sha.* Exact modular-symbol computations with PARI/GP [32] give $L(E, 1) \neq 0$ for both curves. By modularity [37, 6], Gross–Zagier [14] and Kolyvagin [19], both curves have rank 0 and finite Sha. The absolute values $|L(E, 1)|/\Omega_E^+ = 4$ and 16 follow from the primitive integral modular symbol once its scale is fixed; the curves are semistable and optimal, so their Manin constant is 1 by Česnavičius [9, Theorem 1.2].
2. *Local data.* The curve 571a1 has reduction type I_1 at 571; the curve 2045b1 has types I_{17} at 5 (nonsplit) and I_1 at 409. All Tamagawa numbers are 1, and both torsion groups are trivial.
3. *2-descent.* A 2-descent in PARI/GP, with the class groups of the descent fields fully certified (so independent of GRH), gives $\dim \text{Sha}[2] = 2$ for both curves. For 571a1 the Cassels pairing is nondegenerate on $\text{Sha}[2]$. If $x \in \text{Sha}[4]$ then $\langle 2x, y \rangle = \langle x, 2y \rangle = 0$ for all $y \in \text{Sha}[2]$, so $2x = 0$; repeating the argument gives $\text{Sha}[2^\infty] = \text{Sha}[2] \cong H_2$. For 2045b1 the descent shows $\text{Sha}[2] = 2 \text{Sha}[4]$; choosing a half of each element of $\text{Sha}[2]$ gives a bijection $\text{Sha}[2] \times \text{Sha}[2] \rightarrow \text{Sha}[4]$, so $\# \text{Sha}[4] = 16$.
4. *Eight-descent.* For 2045b1 we use the published eight-descent computation in the proof of Miller [21, Theorem 7.1], which applies to optimal curves of conductor below 5000 whose analytic 2-primary prediction has valuation 4; it gives $\text{Sha}[8] = \text{Sha}[4]$. Hence $\text{Sha}[2^\infty] = \text{Sha}[4]$, a group of order 16 and 2-rank 2, that is, H_4 . We use this intermediate computation, not the BSD formula.
5. *Whole orders.* Since Sha is finite, $\# \text{Sha} = 4u$ for 571a1 and $16v$ for 2045b1 with u, v odd. These differ, because $4u \equiv 4$ and $16v \equiv 0 \pmod{8}$; the odd parts are not needed.

Lean checks the group-theoretic steps: the collapse $\text{Sha}[2^\infty] = \text{Sha}[2]$ from nondegeneracy, the halving bijection and the resulting count of $\text{Sha}[4]$, the stabilization from $\text{Sha}[8] = \text{Sha}[4]$, the classification of perfect alternating pairings on $(\mathbb{Z}/4\mathbb{Z})^2$ (each is a unit multiple of the standard one), and the congruence in step 5. For 571a1 it also checks the explicit quartic-cover maps and the numerical local-solubility conditions of the descent. The descent and modular-symbol computations, the arithmetic theorems in steps 1–4, and the published eight-descent are external inputs.

Obligation 5.6 (Finite-source transport). The Bloch–Kato finite component requires a comparison

$$\beta_{\text{BK,finite}} : L_{\text{finite}}(E)_{\mathbb{R}} \longrightarrow \Delta_f(M_E)_{\text{finite}}$$

that fixes the integral local conditions at bad primes, the squared-torsion normalization, and the determinant-line convention for the Sha factor: an integral presentation of each paired group, together with a generator and an orientation of the corresponding determinant line. The paired group determines the order $\#S_p$, but the pairing does not canonically select a rational lift of its Pfaffian coefficient, nor a prescribed generator or orientation of the determinant line; these must come from the presentation. Even an integral presentation does not determine a prescribed generator from the determinant ideal. For the ordinary Selmer complex of the curve 1913b1 at $p = 3$, whose inverse determinant ideal in $\mathbb{Z}_3$ is (9), the elements 9 and 18 both generate that ideal, but only 9 is a square in $\mathbb{Q}_3$; the construction is described in the companion paper [33]. This comparison is external arithmetic input and is not derived from the finite calculations above.

6 Local p -adic column

The local p -adic column records local data at a prime p before they are contracted to a single local number. At a good ordinary prime this means keeping the ordinary filtration together with the finite local condition and its logarithm. At a good supersingular prime it means keeping the signed $+$ and $-$ local conditions separately rather than adding them.

Definition 6.1 (Real p -adic map). Let $T = T_p E$ and $V = V_p E = T \otimes \mathbb{Q}_p$, as representations of $G_{\mathbb{Q}_p}$, and suppose E has good reduction at p . The finite local condition [5, 12] is

$$H_f^1(\mathbb{Q}_p, V) = \ker(H^1(\mathbb{Q}_p, V) \rightarrow H^1(\mathbb{Q}_p, V \otimes B_{\text{cris}})).$$

Since 1 is not an eigenvalue of Frobenius on $D_{\text{cris}}(V)$, the Bloch–Kato exponential $\exp_V : D_{\text{dR}}(V)/\text{Fil}^0 D_{\text{dR}}(V) \rightarrow H_f^1(\mathbb{Q}_p, V)$ is an isomorphism of one-dimensional $\mathbb{Q}_p$ -vector spaces; its inverse is the logarithm

$$\log_V : H_f^1(\mathbb{Q}_p, V) \xrightarrow{\sim} D_{\text{dR}}(V)/\text{Fil}^0 D_{\text{dR}}(V).$$

If E has good ordinary reduction [20], with ordinary filtration $0 \rightarrow T^+ \rightarrow T \rightarrow T^- \rightarrow 0$, the native ordinary family is

$$L_p^{\text{ord}}(E) = (T^+, H_f^1(\mathbb{Q}_p, V), \log_V).$$

If p is odd and E has good supersingular reduction with $a_p = 0$, the native signed family [18, 26] is

$$L_p^{\pm}(E) = (H_+^1, H_-^1, \kappa_{\pm}),$$

where $H_{\pm}^1$ are Kobayashi’s plus and minus local conditions (at a specified level of the cyclotomic tower, with specified coefficients) and $\kappa_{\pm}$ are the corresponding plus and minus Coleman maps. Other supersingular cases, including $p = 2$, are outside this definition.

These native families consist of p -adic objects. A real local map

$$\rho_{p, \text{ord}}^{\mathbb{R}} : L_p^{\text{ord}}(E) \longrightarrow \Delta_f(V, T)_{p, \mathbb{R}}^{\text{ord}}, \quad \rho_{p, \pm}^{\mathbb{R}} : L_p^{\pm}(E) \longrightarrow \Delta_f(V, T)_{p, \mathbb{R}}^{\pm},$$

reading them in a real line, is *not* constructed here. It requires a rational structure on the relevant p -adic determinant lines (for example on the line of $\log_V$), and in the signed case a choice of specialization or localization of the Iwasawa-theoretic Coleman maps together with a normalization. We treat these real readouts as supplied data, part of the local transport of Obligation 6.4. The calculations below only concern which native coordinates such a readout is allowed to see.

The logarithm, not the dual exponential, is the right coordinate on the finite condition. Under the identification $V^*(1) \cong V$ given by the Weil pairing, the dual exponential is a map $\exp^* : H^1(\mathbb{Q}_p, V) \rightarrow \text{Fil}^0 D_{\text{dR}}(V)$ with a different target, and it vanishes on $H_f^1(\mathbb{Q}_p, V)$: by the reciprocity law relating $\exp^*$ to $\exp$ under the local Tate pairing, and the orthogonality of the finite conditions for V and $V^*(1)$, every pairing of $\exp^*(h)$ with $D_{\text{dR}}(V^*(1))$ vanishes for $h \in H_f^1(\mathbb{Q}_p, V)$, so $\exp^*(h) = 0$ [5, 16]. Restricted to the finite condition, the dual exponential therefore carries no information.

Theorem 6.2 (Ordinary local Schur collapse). *For a good ordinary row, let $x = (x_{\text{ord}}, x_f, x_{\log}) \in \mathbb{R}^3$ be tangent coordinates for the ordinary filtration, the finite local condition and the logarithm. With $D_{\text{ord}}(x) = x$, $L_{\text{ord}, \text{full}}(x) = x$ and $C = I_3$, one has $\Xi_C(D \mid L) = 0$. If the coordinate x_{ord} is omitted from the source, the residual has trace 1.*⁸

⁸Lean proves the coordinate calculation. Its formal ordinary local map uses the logarithm, supplied together with its inverse exponential as an explicit isomorphism, and Lean separately derives that the dual exponential vanishes on finite classes from the reciprocity and orthogonality laws. Lean does not show that the three coordinates vary independently for an actual curve, and does not construct the local determinant comparison.

Proof. With the full source all blocks equal I_3 , so $\Xi_C(D \mid L) = I_3 - I_3 I_3 I_3 = 0$. Omitting x_{ord} leaves the source spanned by $(0, 1, 0)$ and $(0, 0, 1)$, with projector $\text{diag}(0, 1, 1)$; the residual $\text{diag}(1, 0, 0)$ has trace 1. $\square$

Theorem 6.3 (Signed local Schur collapse). *For a signed row as in Definition 6.1 (p odd, good supersingular reduction, $a_p = 0$), let $y = (y_+, y_-, y_\kappa) \in \mathbb{R}^3$ be tangent coordinates for the two signed conditions and the signed comparison. With $D_\pm(y) = y$, $L_{\pm, \text{full}}(y) = y$ and $C = I_3$, one has $\Xi_C(D \mid L) = 0$. The unsigned source $L_{\text{uns}}(y) = (y_+ + y_-, y_\kappa)$ has rank 2, leaves the direction $(1, -1, 0)$ invisible, and has residual trace 1.⁹*

Proof. With the full source, $\Xi_C(D \mid L) = 0$ as in the ordinary case. The unsigned source is spanned by the orthogonal vectors $u_1 = (1, 1, 0)$ and $u_2 = (0, 0, 1)$, so its projector is $P = u_1 u_1^\top / 2 + u_2 u_2^\top$, of trace $2/2 + 1 = 2$, and $\text{tr}(I_3 - P) = 1$. The vector $(1, -1, 0)$ is orthogonal to u_1 and u_2 ; equivalently, the signed inputs $(1, 0, 0)$ and $(0, 1, 0)$ have the same unsigned image $(1, 0)$. $\square$

Obligation 6.4 (Local p -adic transport). The Bloch–Kato local component requires a comparison

$$\beta_{\text{BK}, p} : \Delta_f(V, T)_{p, \mathbb{R}} \longrightarrow \Delta_f(M_E)_p$$

together with the real readouts $\rho_{p, \text{ord}}^{\mathbb{R}}$ and $\rho_{p, \pm}^{\mathbb{R}}$ of Definition 6.1, fixing the rational structures and normalizations of the p -adic determinant lines, the integral Fontaine–Laffaille or Wach-module data, the orientation of the logarithm and of the local Tate pairing [22], local–global compatibility of Selmer complexes [24], the signed normalization at supersingular primes, and compatibility with the determinant-assembly column. This comparison is external arithmetic input; the coordinate calculations above do not construct it.

7 Determinant-assembly column

The four arithmetic columns produce four factors. To obtain an element of one determinant line one also needs an orientation: the signs, units and trivialization conventions under which the factors are multiplied. The determinant-assembly column holds this orientation as an independent input.

Definition 7.1 (Real determinant-assembly map). Let

$$\lambda_{\text{loc}}(E) = \rho_{\text{ht/reg}}^{\mathbb{R}} \otimes \rho_{\text{an/period}}^{\mathbb{R}} \otimes \rho_{\text{finite}}^{\mathbb{R}} \otimes \bigotimes_{p \in S(E)} \rho_p^{\mathbb{R}}$$

be the tensor of the four arithmetic columns over a finite support set $S(E)$ of primes, with the local factors at primes outside $S(E)$ trivialized as units. Let $o_{\text{det}} \in C_{\text{det}}(E)$ be an orientation and trivialization datum, supplied independently of the four columns, and let $\rho_{\text{det}}^{\mathbb{R}}(o_{\text{det}}) \in \mathbb{R}^\times$ be the orientation scalar it determines (the map on the orientation slot from Definition 2.1). The real determinant assembly is

$$\mathcal{A}^{\mathbb{R}}(E) = \lambda_{\text{loc}}(E) \cdot \rho_{\text{det}}^{\mathbb{R}}(o_{\text{det}}) \in \Delta_f(M_E)_{\mathbb{R}}.$$

We call $\mathcal{A}^{\mathbb{R}}(E)$ the *final assembly*. It already contains the four arithmetic factors, and so must not be used again as a factor alongside them; see Definition 8.2.

⁹Lean proves the coordinate calculation over $\mathbb{Q}$ and exhibits the two signed inputs with the same unsigned image. The signed comparison is a separately supplied carrier; actual signed local conditions, regulators and determinants are not constructed.

The adequacy question for this column is whether the four arithmetic columns, without the orientation, can explain the assembled target. In the coordinate model they cannot, and the shortfall is exactly one direction.

Theorem 7.2 (Cascade-completion Schur signature). *Use the tangent coordinates*

$$x = (x_{\text{ht}}, x_{\text{an}}, x_{\text{fin}}, x_p, x_{\text{det}}) \in \mathbb{R}^5.$$

Let the four arithmetic columns supply the source row $\ell = (1, 1, 1, 1, 0)$ and let the assembled target have row $d = (1, 1, 1, 1, 1)$. With $C = I_5$,

$$K_{DD} = dd^\top = 5, \quad K_{LL} = \ell\ell^\top = 4, \quad K_{DL} = d\ell^\top = 4, \quad \Xi_C(D \mid L) = 5 - 4 \cdot 4^{-1} \cdot 4 = 1.$$

The residual is the orientation direction $e_{\text{det}} = (0, 0, 0, 0, 1)$.¹⁰

Proof. The three blocks are the displayed sums of squares and products of entries, and the one-dimensional source block 4 has inverse 4^{-1} . Hence $\Xi_C(D \mid L) = 5 - 4 = 1$. Since $d = \ell + e_{\text{det}}$ with $\ell \cdot e_{\text{det}} = 0$, projecting d onto ℓ leaves e_{det} , of squared length 1. $\square$

Obligation 7.3 (Integral determinant assembly). The full determinant comparison requires, for each orientation datum o_{det} , a transport

$$\beta_{\text{det-assembly}}(\cdot; o_{\text{det}}) : \Delta_{\text{ht/reg}} \otimes \Delta_{\text{an/period}} \otimes \Delta_{\text{finite}} \otimes \bigotimes_{p \in S(E)} \Delta_p \longrightarrow \Delta_f(M_E),$$

together with compatible unit trivializations of the local lines at primes outside $S(E)$. It must fix the Knudsen–Mumford determinant signs [17], the local–global compatibility of Selmer complexes, the convention relating the zeta element to the motivic trivialization, and compatibility with the four column transports. The orientation datum is a parameter of the transport, distinct from the assembled output. This transport is external arithmetic input. The residual computation above shows only that, in the coordinate model, the four arithmetic rows do not account for the orientation direction; it does not show that the orientation varies independently for actual curves.

8 Comparison with the Bloch–Kato scalar package

The preceding sections built the five columns separately. This section contracts them back to the usual Bloch–Kato scalar package and asks two questions: is the typed source adequate for that contraction, and what does the contraction forget?

Remark 8.1 (Relation to the standard presentations). The fundamental line and its Tamagawa-number formulation go back to Bloch–Kato [5, §5] and Fontaine–Perrin-Riou [13]; the formulation of the leading value in terms of periods and height determinants, modulo $\mathbb{Q}^\times$, is surveyed by Nekovář [23]. The five-slot organization used here is a bookkeeping choice made for the adequacy tests. We have not compared it systematically with the literature and make no claim about whether equivalent decompositions appear there.

Definition 8.2 (Five-column tensor). The typed five-column tensor is

$$T_{5\text{col}}(E) = \rho_{\text{ht/reg}}^{\mathbb{R}}(E) \otimes \rho_{\text{an/period}}^{\mathbb{R}}(E) \otimes \rho_{\text{finite}}^{\mathbb{R}}(E) \otimes \bigotimes_{p \in S(E)} \rho_p^{\mathbb{R}}(E) \otimes \rho_{\text{det}}^{\mathbb{R}}(o_{\text{det}}),$$

¹⁰Lean proves these exact integer values. The statement concerns the orientation direction only; it does not assert adequacy of each of the four arithmetic coordinates.

whose fifth factor is the orientation scalar of the independent orientation datum (Definition 7.1), not the final assembly $\mathcal{A}^{\mathbb{R}}(E)$. Its source family consists of the five separated logarithmic tangent coordinates $(x_{\text{ht}}, x_{\text{an}}, x_{\text{fin}}, x_p, x_{\text{det}})$, where $x_{\text{det}} = \log |\rho_{\text{det}}^{\mathbb{R}}(o_{\text{det}})|$. The *orientation-normalized scalar readout* is

$$\log |T_{5\text{col}}(E)/\rho_{\text{det}}^{\mathbb{R}}(o_{\text{det}})| = x_{\text{ht}} + x_{\text{an}} + x_{\text{fin}} + x_p,$$

the logarithm of the absolute value of the product of the four arithmetic factors. It is distinct from the full contraction $\log |T_{5\text{col}}(E)|$, whose row is $(1, 1, 1, 1, 1)$.

The distinction in the definition matters. Since $\mathcal{A}^{\mathbb{R}}(E)$ already contains the four arithmetic factors, using it as the fifth factor would count each of them twice. For instance, with arithmetic factors 2, 3, 5, 7 and orientation scalar 1, the final assembly is 210; the tensor with the orientation as fifth factor is again 210, whereas reusing the final assembly would give $210^2 = 44100$. (Lean checks these values.)

Theorem 8.3 (Five-column to Bloch–Kato scalar contraction). *In the coordinates $x = (x_{\text{ht}}, x_{\text{an}}, x_{\text{fin}}, x_p, x_{\text{det}}) \in \mathbb{R}^5$, the orientation-normalized scalar readout of Definition 8.2 is $\Phi_{5\text{col} \rightarrow \text{BK}}(x) = d \cdot x$ with $d = (1, 1, 1, 1, 0)$. With the full typed source $L(x) = x$ and $C = I_5$,*

$$\Xi_{5\text{col} \rightarrow \text{BK}} = \|d\|^2 - d I_5 d^{\top} = 4 - 4 = 0,$$

so the typed source is adequate for this readout. The orientation direction $e_{\text{det}} = (0, 0, 0, 0, 1)$ lies in the kernel of $\Phi_{5\text{col} \rightarrow \text{BK}}$: the readout forgets at least that coordinate.¹¹

Proof. With the identity source, $K_{LL} = K_{LL}^{\dagger} = I_5$, $K_{DL} = d$ and $K_{LD} = d^{\top}$, so the projected energy is $d I_5 d^{\top} = dd^{\top} = 4$, which equals K_{DD} ; hence the residual is 0. Finally $d \cdot e_{\text{det}} = 0$, so changing only x_{det} changes the typed coordinate without changing the scalar readout. $\square$

The two halves of the theorem fit together. The orientation-normalized readout is computed correctly from the typed source, so nothing is lost in that direction; but the typed source carries an orientation coordinate that this readout does not see, while the full tensor does (changing the orientation scalar from 1 to 2 in the example above changes the tensor from 210 to 420). To the extent that the arithmetic Bloch–Kato scalar is orientation-normalized in this sense, which is an additional convention not supplied by the transports alone, it is a projection of the typed atlas.

9 Rank two: the height matrix and its sources

In rank at least 2 the height/regulator column raises a question that does not arise in rank 1. Once a Mordell–Weil basis is known, the Néron–Tate Gram matrix computes the regulator. But a theory that explains the regulator through independent cycle classes (generalized Gross–Zagier [14], Beilinson–Kato [1, 16], $K_2(E)$ or higher Chow groups [4]) would need more than the Gram matrix records. This section makes the rank-two height calculation explicit and then states, in the simplest form, what the Gram matrix forgets.

Definition 9.1 (Rank-two Mordell–Weil/Kummer source). Let $\Lambda_E = E(\mathbb{Q})/E(\mathbb{Q})_{\text{tors}}$ and, for a prime p , let

$$\kappa_p : \Lambda_E \otimes \mathbb{Q}_p \longrightarrow H_f^1(\mathbb{Q}, V_p E)$$

¹¹Lean proves the contraction row, the zero residual and the kernel vector. Identifying this coordinate readout with the arithmetic Bloch–Kato scalar requires the five transports of the column sections and, in addition, a convention under which that scalar is independent of the orientation datum.

be the Kummer map into the Bloch–Kato Selmer group. For a curve of rank 2 with a fixed Mordell–Weil basis (P_1, P_2) , the rank-two Mordell–Weil/Kummer source is the pair $(\kappa_p(P_1), \kappa_p(P_2))$. It is a classical Selmer source once the basis is known; it is not, by this definition, a construction of Heegner cycles or other motivic classes.

For the curve 389a1, $y^2 + y = x^3 + x^2 - 2x$, of rank 2, take $P_1 = (0, 0)$ and $P_2 = (1, 0)$. Computing Néron–Tate heights with PARI/GP [32] gives the Gram matrix

$$H = \begin{pmatrix} 0.3270008\dots & 0.0585227\dots \\ 0.0585227\dots & 0.4767117\dots \end{pmatrix}, \quad \det H = 0.1524602\dots$$

This determinant equals the regulator recorded in the LMFDB [31]. Since a subgroup of index m in Λ_E has Gram determinant $m^2 \text{Reg}(E/\mathbb{Q})$, the points P_1, P_2 form a basis of Λ_E . By Theorem 3.2, the row of $d(\log \det)_H$ in the coordinates $(\delta h_{11}, \delta h_{12}, \delta h_{22})$ is

$$\ell = \left(\frac{h_{22}}{\det H}, -\frac{2h_{12}}{\det H}, \frac{h_{11}}{\det H} \right) = (3.12679\dots, -0.76771\dots, 2.14483\dots), \quad \|\ell\|^2 = 14.96650\dots$$

Theorem 9.2 (Rank-two height Schur collapse). *For the row ℓ above, with the full symmetric height source and $C = I_3$,*

$$\Xi_{\text{NT} \rightarrow \text{Reg}} = \|\ell\|^2 - \ell I_3 \ell^T = 0.$$

The source consisting of the single direction ℓ (the “determinant-only” source) has projector of trace 1 and leaves a residual of trace $3 - 1 = 2$.¹²

Proof. With the full source, $K_{LL} = K_{LL}^\dagger = I_3$ and $\ell I_3 \ell^T = \|\ell\|^2 = K_{DD}$, so the residual is 0. The orthogonal projector onto the line spanned by $\ell \neq 0$ is $P_\ell = \ell^T \ell / \|\ell\|^2$, of trace 1, so $\text{tr}(I_3 - P_\ell) = 2$. $\square$

The determinant is therefore an adequate readout of itself, but it does not retain the three-dimensional height matrix. The next statement goes one step further: even the full height matrix does not retain data that a cycle-theoretic explanation of the regulator would have to supply.

Theorem 9.3 (The Gram matrix forgets provenance data). *Fix $E/\mathbb{Q}$ of Mordell–Weil rank $r \geq 2$ and a set $\mathcal{P}$ of provenance data with at least two elements (for example, records of which cycle construction is claimed to produce each basis class). Define a cycle-source record to be a Mordell–Weil basis $(P_1, \dots, P_r)$ of Λ_E together with an element of $\mathcal{P}$. The map sending a record to its Gram matrix $(\langle P_i, P_j \rangle_{\text{NT}})_{i,j}$ is not injective. In particular, the Gram matrix, which determines $\text{Reg}(E/\mathbb{Q}) = \det(\langle P_i, P_j \rangle_{\text{NT}})$, does not determine the provenance data.¹³*

Proof. Since $r = \text{rank } E(\mathbb{Q})$, a basis exists. Fix one and attach to it two distinct elements of $\mathcal{P}$. The two records are distinct and have the same Gram matrix, because the Gram matrix depends only on the points. $\square$

The theorem is deliberately weak: it says only that the height matrix, which is a function of the points, cannot certify anything about where those points come from. It does not rule out cycle constructions such as those of Bertolini–Darmon–Prasanna [2], Beilinson–Kato, or higher Chow groups, which bring their own input. In the companion paper [33] the rank- ≥ 2 leading-term identity enters as a supplied hypothesis, not as a theorem.

¹²Lean proves the statement for the integer row $(312679, -76771, 214483)$, which is $10^5 \ell$ rounded to integers, and proves the trace-one projector onto that scaled direction over $\mathbb{Q}$. The general rank-two row formula is proved separately (Theorem 3.2); the numerical heights of 389a1 are computed outside Lean.

¹³Lean proves a narrower record-model statement: the projection from records consisting of an integer Gram identifier and three Boolean metadata fields to the identifier is not injective. It constructs no Mordell–Weil bases, height matrices or cycle classes.

10 Finite windows of curves

Numerical checks of BSD-type statements are made on finite lists of curves. This section records the elementary but easily forgotten fact that a statement about all curves cannot be decided from a fixed finite list. The list used for examples in this paper is the five-curve window

$$W = \{11a1, 37a1, 121b1, 960d1, 571a1\}.$$

Theorem 10.1 (A finite window does not decide a universal statement). *Let $\mathcal{C}$ be the set of isomorphism classes of elliptic curves over $\mathbb{Q}$, and consider assignments $\tau : \mathcal{C} \rightarrow \{\text{true}, \text{false}\}$, thought of as truth values of some property of curves. Let $U(\tau)$ be the statement “ $\tau(E) = \text{true}$ for every E ”.*

1. *No function of the restriction $\tau|_W$ computes $U(\tau)$ for all assignments τ : the assignment $\tau_1 \equiv \text{true}$ and the assignment τ_2 that agrees with τ_1 except at one curve $E_* \notin W$, where it is false, have the same restriction to W , but $U(\tau_1)$ is true and $U(\tau_2)$ is false.*
2. *In Schur form, let $\mathcal{C}(B)$ be the finite set of curves of conductor at most $B \geq 960$, with $M(B) = \#\mathcal{C}(B)$ coordinates. The window source is the coordinate projection onto the five coordinates in W , and the residual against the identity readout on $\mathcal{C}(B)$ has trace $M(B) - 5$, which tends to infinity with B .¹⁴*

Proof. (1) Any function of $\tau|_W$ takes the same value on τ_1 and τ_2 , while U separates them. (2) The trace of a coordinate projection is the number of retained coordinates, so $\text{tr}(I_{M(B)} - P_W) = M(B) - 5$. There are infinitely many isomorphism classes of elliptic curves over $\mathbb{Q}$ and finitely many of each conductor, so $M(B) \rightarrow \infty$. $\square$

The point of the statement is methodological. A conclusion about all curves needs an argument that reaches the curves outside the window, for instance a theorem covering a family, or an exhaustion argument, or a passage from density statements to pointwise ones. The finite window by itself supplies none of these.

11 Finite sets of primes

The local columns are indexed by primes. This section records the local analogue of the previous one: a fixed finite set of primes cannot certify a statement about all primes. It then describes the numerical support table used for the examples, and says precisely what that table does and does not establish.

Theorem 11.1 (A finite set of primes does not decide all-prime coverage). *Let a coverage record assign to each prime p a status, covered or uncovered, and let the all-prime target be the statement that every prime is covered. For every finite set P_0 of primes there is a prime $q \notin P_0$, and there are two coverage records that agree at every prime of P_0 , such that the first covers every prime and the second leaves q uncovered. Consequently no function of the restriction of a record to P_0 decides the all-prime target.¹⁵*

¹⁴Lean proves both parts for a model in which curves are indexed by natural numbers and the window is the first five indices: the two completions agree on the window, differ outside it, are separated by the universal target, and are identified by every function of the window; and the residual trace $M - 5$ takes every natural-number value. The model says nothing about which truth assignments arise from actual properties of curves.

¹⁵Lean proves the statement for coverage records indexed by natural numbers, with an explicit prime q dividing $1 + \prod_{p \in P_0} p$, and with primality decided by a finite divisor check. These are formal records; no claim is made that they arise from elliptic curves.

Proof. Let q be a prime factor of $1 + \prod_{p \in P_0} p$; then $q \notin P_0$. Let the first record cover every prime and let the second agree with it except at q . The two records have the same restriction to P_0 , so any function of that restriction gives them the same value, while the all-prime target holds for the first and fails for the second. $\square$

For computations one works with a finite truncation of the local column.

Definition 11.2 (Support-prime truncation residual). Fix a curve E and a set $R(E)$ of primes, the *residual rows*: the primes at which no local comparison is taken as available. For $B \geq 1$ the truncation residual is

$$\Xi_B(E) = \#\{p \leq B : p \text{ prime}, p \in R(E)\}.$$

It depends only on $R(E)$ below B . In the examples below $R(E)$ is the set of primes of bad reduction. For the five window curves and $B = 49$ this gives

$$\Xi_{49}(11a1) = \Xi_{49}(37a1) = \Xi_{49}(121b1) = 1, \quad \Xi_{49}(960d1) = 3, \quad \Xi_{49}(571a1) = 0,$$

the last becoming 1 once $B \geq 571$. This is a finite diagnostic, distinct from Theorem 11.1.

Taking $R(E)$ to be the bad primes treats every good prime as covered by an ordinary or signed local theory. For good ordinary primes this is the setting of Theorem 6.2. For good supersingular primes, the signed theory of Definition 6.1 requires p odd and $a_p = 0$, and the table below shows that this fails in some small cases.

Remark 11.3 (Numerical support table). For the five window curves and the 15 primes below 50, together with the pair $(571a1, 571)$, we have computed a 76-row table of (E, p) data from the Cremona models: the number of points of the reduction and a_p (each obtained by two independent methods), the reduction type (good, multiplicative or additive), and for good primes whether the reduction is ordinary or supersingular. The table is included with the supporting files of the paper. The bad primes are

$$11a1 : \{11\}, \quad 37a1 : \{37\}, \quad 121b1 : \{11\}, \quad 960d1 : \{2, 3, 5\}, \quad 571a1 : \{571\}.$$

Among the good supersingular rows, five lie outside the basic numerical scope (p odd and $a_p = 0$) of Kobayashi's signed theory [18]: $(11a1, 2)$ and $(37a1, 2)$ with $a_2 = -2$; $(37a1, 3)$ with $a_3 = -3$; and $(121b1, 2)$ and $(571a1, 2)$ with $a_2 = 0$. For these rows the cited signed input does not supply a local comparison. Generalized signed constructions that allow $p = 2$ or $a_p \neq 0$ exist, but would have to be specified and checked row by row. If these five rows are counted as residual, the truncation residuals below 50 become 2, 3, 2, 3, 1 for $11a1, 37a1, 121b1, 960d1, 571a1$. The table certifies the numerical local data only; it does not certify any local comparison or main-conjecture input for any row.

The finiteness statements of this paper are collected in Table 2.

12 A formal source algebra in rank two

The rank-two discussion of Section 9 suggests separating two questions: what algebraic shape a rank- r source should have, and whether actual cycle classes realize that shape. This section treats the first question with the simplest possible device, the exterior algebra on r formal symbols. The top product $J_1 \cdots J_r$ plays the role of a determinant: it is nonzero exactly when the symbols are independent. We call this algebra the *virtual source algebra* vsrc_r ; the word “virtual” signals that the symbols are not, by themselves, cycle classes.

Table 2. The “does not determine” results of the paper. Each concerns a specific readout on a specific domain; none excludes richer source data. The last column says on what objects the statement is proved.

Readout	What it does not determine	Proved on
Valuation of the scalar finite factor (Theorems 2.4 and 5.3)	whether a valuation sits in the Sha block or the Tamagawa block; the Cassels–Tate datum	formal finite-source records and coordinates
Dimension of Sha[2] (Theorem 5.4, Example 5.5)	the order of the 2-primary paired group, and of Sha	finite paired groups; arithmetically, the curves 571a1 and 2045b1, using external descent inputs
Rank-two regulator determinant (Theorem 9.2)	the three-dimensional height matrix	coordinates, with numerical heights of 389a1
Gram matrix in rank ≥ 2 (Theorem 9.3)	provenance data of the basis classes	abstract records
A finite window of curves (Theorem 10.1)	a statement about all curves	arbitrary truth assignments
A finite set of primes (Theorem 11.1)	coverage of all primes	arbitrary coverage records

Definition 12.1 (Virtual source algebra). Let $V_r = \mathbb{Q}J_1 \oplus \cdots \oplus \mathbb{Q}J_r$ and $\text{vsr}_r = \Lambda_{\mathbb{Q}}(V_r)$, the exterior algebra. In rank 2 it has ordered basis 1, J_1 , J_2 , $J_{12} := J_1J_2$ and relations

$$J_1^2 = 0, \quad J_2^2 = 0, \quad J_1J_2 = J_{12}, \quad J_2J_1 = -J_{12}.$$

The element J_{12} is the determinant-shaped witness of independence of J_1 and J_2 .¹⁶

Figure 2 displays the rank-two algebra.

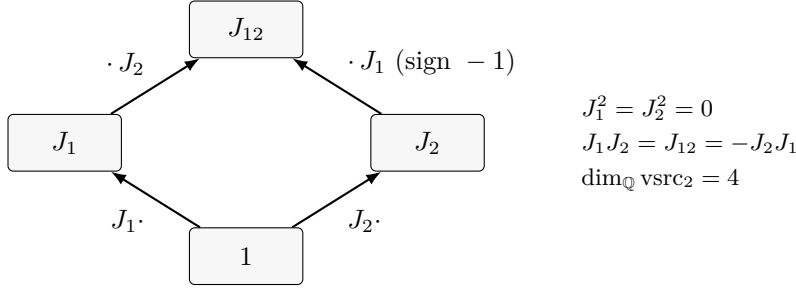


Figure 2. The rank-two virtual source algebra $\text{vsr}_2 = \Lambda_{\mathbb{Q}}(\mathbb{Q}J_1 \oplus \mathbb{Q}J_2)$. Basis elements correspond to subsets of $\{1, 2\}$; multiplication by a symbol moves up one level, and the top element J_{12} records the independence of the two symbols.

Theorem 12.2 (Stage I: consistency). *vsr₂ is an associative, unital, graded-commutative $\mathbb{Q}$ -algebra. Every element has a unique normal form $a_{\emptyset}1 + a_1J_1 + a_2J_2 + a_{12}J_{12}$ with $a_{\bullet} \in \mathbb{Q}$, so $\dim_{\mathbb{Q}} \text{vsr}_2 = 4$. In lower rank, $\text{vsr}_0 = \mathbb{Q}$ and $\text{vsr}_1 = \mathbb{Q} \oplus \mathbb{Q}J_1$ with $J_1^2 = 0$.*¹⁷

¹⁶Lean constructs the rank-two multiplication table on integer coordinates and proves associativity, the unit laws and additive and scalar bilinearity. Rational coefficients, other ranks and the dimension count are treated here on paper.

¹⁷Lean proves the unit laws, the four relations and the distinctness of the basis symbols for the integer-coordinate model, and records the numbers $2^0, 2^1, 2^2$; it does not prove the dimension statement over $\mathbb{Q}$.

Proof. Take the $\mathbb{Q}$ -vector space with basis indexed by the subsets of $\{1, 2\}$. Define the product of two basis vectors to be 0 if the subsets meet, and otherwise to be the basis vector of the union times the sign of the permutation that sorts the concatenated index list; extend bilinearly. This gives the displayed relations, and the empty subset is a unit. Associativity holds on basis vectors because both parenthesizations vanish when an index repeats, and otherwise both produce the same union with the sign of the same sorting permutation. Moving a degree- a monomial past a degree- b monomial costs $(-1)^{ab}$, which is graded commutativity. The four subsets give a basis, so the dimension is 4; the same construction with $\{1\}$ or $\emptyset$ gives vsrc_1 and vsrc_0 . $\square$

Remark 12.3 (No rank- r realization is supplied here). For $r \geq 2$, nothing in this paper provides a map

$$\text{vsrc}_r(E) \longrightarrow \text{CH}^*(\mathcal{X})_{\mathbb{Q}},$$

or into a Selmer or Mordell–Weil group, that realizes $J_1, \dots, J_r$ as independent cycle classes. The Gram-matrix statement Theorem 9.3 shows that heights alone do not supply such provenance. Constructions of cycle classes with arithmetic meaning, such as Kato’s Euler system [16] and generalized Heegner cycles [2], come with their own hypotheses; we do not use them, and we make no claim about whether they realize vsrc_r for a given curve.

In rank 1 the situation is classical. Under the Gross–Zagier hypotheses [14, 19] for an imaginary quadratic field K , there is a Heegner point $P_K \in E(K)$, and one can send J_1 to P_K .

Theorem 12.4 (Stage II: rank one). *Send $J_1 \mapsto P_K$ and the formal height $h_{\text{vsrc}}(J_1, J_1)$ to the Néron–Tate height $\langle P_K, P_K \rangle_{\text{NT}}$.*

1. *The 1×1 Gram determinant of the image source is $\langle P_K, P_K \rangle_{\text{NT}}$. If P_K has infinite order and $E(K)$ has rank 1, this equals $[\Lambda : \mathbb{Z}P_K]^2 \text{Reg}(E/K)$, where $\Lambda = E(K)/E(K)_{\text{tors}}$ and $[\Lambda : \mathbb{Z}P_K]$ is the index of the image of $\mathbb{Z}P_K$.*
2. *The Gross–Zagier identity relating $L'(E/K, 1)$ to $\langle P_K, P_K \rangle_{\text{NT}}$ is not determined by these symbolic data: two formal data sets can share P_K and $\langle P_K, P_K \rangle_{\text{NT}}$ while having different analytic quotients. Relative to the symbolic calculus the analytic identity is an additional input, which we record by a nonzero residual Ξ_{GZ} .¹⁸*

Proof. (1) The Gram determinant of a single vector is its height. If Q generates Λ and $P_K \equiv mQ$ modulo torsion, then $\langle P_K, P_K \rangle_{\text{NT}} = m^2 \langle Q, Q \rangle_{\text{NT}} = m^2 \text{Reg}(E/K)$ with $m = [\Lambda : \mathbb{Z}P_K]$. (2) The symbolic data contain neither $L'(E/K, 1)$ nor the period, so they cannot constrain the analytic side; the identity is the content of the Gross–Zagier theorem. $\square$

In rank 2 the corresponding statement is a one-way translation. For 389a1 the Tamagawa product and the torsion order are 1; we write

$$\Xi_{\text{Sc}} = A_E - \Omega_E^+ \text{Reg}(E/\mathbb{Q}) \# \text{Sha}(E/\mathbb{Q})$$

for the scalar BSD residual, assuming $\text{Sha}(E/\mathbb{Q})$ finite, which is not known for this curve. A *rank-two realization* consists of a $\mathbb{Q}$ -vector space W of classes (for instance cycle classes modulo homological equivalence), a linear map $f : V_2 \rightarrow W$, a symmetric bilinear height pairing on W , and a bridge $\mathcal{B}_{\text{an-ht}}^{(2)}$ identifying A_E with $\Omega_E^+ \# \text{Sha}$ times the Gram determinant of $(f(J_1), f(J_2))$ for that pairing. Put $r_2 = \Lambda f : \text{vsrc}_2 \rightarrow \Lambda_{\mathbb{Q}}(W)$; then $z_{12} = r_2(J_{12}) = f(J_1) \wedge f(J_2)$, and $z_{12} \neq 0$ exactly

¹⁸Lean checks the symbolic assignments $J_1 \mapsto P_K$ and $h_{\text{vsrc}}(J_1, J_1) \mapsto (\text{Néron–Tate height})$, and records the declared residual value $\Xi_{\text{GZ}} = 1$; it constructs neither the Heegner point nor any height.

when $f(J_1), f(J_2)$ are linearly independent. Let Ξ_{Mot} be a *motivic residual*, which by definition vanishes exactly when a rank-two realization with $z_{12} \neq 0$ is given. All of these data are supplied; none is constructed here.

Theorem 12.5 (Stage III: one-way translation for 389a1). *For the fixed pair $(\Xi_{\text{Mot}}, \Xi_{\text{Sc}})$ just defined, suppose given the analytic-height bridge in the form of the implication $\Xi_{\text{Mot}} = 0 \Rightarrow \Xi_{\text{Sc}} = 0$. Then a rank-two realization yields the scalar identity $A_E = \Omega_E^+ \text{Reg}(E/\mathbb{Q}) \# \text{Sha}(E/\mathbb{Q})$. The reverse implication $\Xi_{\text{Sc}} = 0 \Rightarrow \Xi_{\text{Mot}} = 0$ is not provided by the scalar data; it holds only if supplied as separate content.*¹⁹

Proof. The forward statement is the supplied implication. No reverse construction of a realization or of an independence certificate from the scalar equality is supplied here. (Theorem 9.3 establishes only that provenance data cannot be recovered from the Gram matrix in its record model; it does not exclude a separately defined construction.) $\square$

The statement must be read for the fixed pair. A universal version, asserting $\Xi_{\text{Mot}} = 0 \Rightarrow \Xi_{\text{Sc}} = 0$ for all values of the two residuals, would be false, for instance at $(\Xi_{\text{Mot}}, \Xi_{\text{Sc}}) = (0, 1)$.

Remark 12.6 (The source reading carries more data). The scalar leading-coefficient formula records a number. A rank- r source statement in the sense above also asks for a nonzero independent wedge $J_1 \wedge \cdots \wedge J_r$ realized by actual classes. In this sense the source reading carries more data than the scalar equality. We do not claim a general theorem that scalar BSD is strictly weaker than any such source statement; the comparison above is for the fixed pair and the supplied bridge.

13 The normalization constant κ_r

Some formulations of the leading term, including the one used in the companion paper, write the arithmetic side as $\kappa_r(E) \Omega_E \text{Reg}_r(E) c_E$ with an unspecified constant $\kappa_r(E)$. This section identifies the constant for which that form is the standard Strong BSD formula, and makes the hypotheses of that identification explicit. It is a comparison of two ways of writing the same conjectural identity, not a derivation of either.

Throughout, $\text{Sha}(E/\mathbb{Q})$ is assumed finite, Ω_E is the BSD real period, $\text{Reg}_r(E)$ is the regulator (1 in rank 0), $\text{Tam}(E) = \prod_\ell c_\ell(E)$, and

$$C = \Omega_E \text{Reg}_r(E) \text{Tam}(E), \quad d = \#E(\mathbb{Q})_{\text{tors}}^2.$$

Theorem 13.1 (Normalization of κ_r). *Fix the convention $c_E = \text{Tam}(E)$ and work in a field containing all the quantities above (for curves, $\mathbb{R}$). Assume $C \neq 0$ and $d \neq 0$.*

1. *The right-hand side of the cascade form*

$$\frac{L^{(r)}(E, 1)}{r!} = \kappa_r(E) \Omega_E \text{Reg}_r(E) c_E$$

equals the right-hand side of the standard Strong BSD form

$$\frac{L^{(r)}(E, 1)}{r!} = \frac{\# \text{Sha}(E/\mathbb{Q}) \text{Reg}_r(E) \Omega_E \text{Tam}(E)}{d}$$

¹⁹The formalization records this as a conditional schema: for a fixed pair of residuals, the forward implication is projected from the supplied bridge, and the reverse from supplied external content. Lean does not construct a realization or a bridge.

if and only if

$$\kappa_r(E) = \frac{\# \text{Sha}(E/\mathbb{Q})}{\# E(\mathbb{Q})_{\text{tors}}^2} \quad \left(\text{equivalently, } \kappa_r(E) d = \# \text{Sha}(E/\mathbb{Q}) \right).$$

2. If this normalization holds, the two forms are equivalent for every value of the left-hand side. Conversely, if the two forms are equivalent for every value of the left-hand side, the normalization holds.

For actual curves the hypotheses hold: $\Omega_E > 0$, the regulator is positive because the Néron–Tate pairing is positive definite on $E(\mathbb{Q}) \otimes \mathbb{R}$, $\text{Tam}(E)$ is a positive integer, and $d \geq 1$.²⁰

Proof. The standard right-hand side equals $(\# \text{Sha} / d) C$, by commutativity and the definition of division. The cascade right-hand side is $\kappa_r(E) C$. Since $C \neq 0$, $\kappa_r(E) C = (\# \text{Sha} / d) C$ if and only if $\kappa_r(E) = \# \text{Sha} / d$; since $d \neq 0$, this holds if and only if $\kappa_r(E) d = \# \text{Sha}$. This proves (1). For (2), equal right-hand sides make the two equations equivalent for every left-hand side. Conversely, evaluate both forms at the left-hand side $\kappa_r(E) C$: the cascade form holds there, so the standard form does too, and (1) gives the normalization. $\square$

Part (2) needs the quantifier over all values of the left-hand side. Equivalence of the two equations at a single, fixed value says nothing: for instance the equations $0 = 1$ and $0 = 2$ are both false, hence logically equivalent, although $1 \neq 2$. When we say below that the cascade form “agrees” with the standard form we mean that their right-hand sides are equal, as in (1).

The conditional closure of the companion paper [33] uses this normalization to read its rank- ≥ 2 hypothesis as the standard Strong BSD formula.

14 Scope

We collect here what the paper does not claim.

1. *No result about the truth of BSD.* The paper proves neither BSD, nor the Bloch–Kato conjecture, nor an Iwasawa main conjecture, nor a Gross–Zagier formula. All statements about adequacy and information loss concern specified readouts.
2. *“Does not determine” is not “does not exist”.* Each result of that form (Table 2) says that a particular readout, on a particular domain, does not determine a particular target. None says that richer data, or a future theorem, cannot recover the target. Several of these statements are proved on formal models (records with tags, truth assignments, coverage records); where an arithmetic realization is available, as for $\text{Sha}[2]$ in Example 5.5, it is stated separately with its external inputs.
3. *Finite coordinate calculations.* The adequacy collapses are finite linear-algebra computations in declared coordinates (identity-matrix actions and projector traces). They are not statements about operators on infinite-dimensional spaces, and they do not show that the declared coordinates vary independently for actual curves.

²⁰The declaration recorded for this theorem is Lean’s cancellation of a nonzero common factor over the integers. Additional Lean declarations prove (1) and (2) over an arbitrary field, including the reordering of the standard product and the cross-multiplied form, which requires $d \neq 0$. Positivity of the period and the regulator for actual curves is not formalized.

4. *The five transports are inputs.* The comparisons from the five native columns to the Bloch–Kato line, including the real readouts of the p -adic data and the orientation convention, are external arithmetic inputs, recorded as obligations in the column sections.
5. *The source algebra is formal.* The algebra vsr_r fixes the shape of a rank- r source; it is not a construction of cycle classes.

15 Discussion

The standard BSD formula is a product of scalar factors: an analytic leading coefficient divided by a period, a regulator, and a finite factor built from Tamagawa numbers, torsion and Sha, with local p -adic normalizations and determinant conventions hidden in the assembly. The formula is a powerful summary, but each factor is the image of richer data. The question of this paper is which of those data a given readout retains.

The Schur residual $\Xi_C(D \mid L)$ turns that question into finite linear algebra. If all native coordinates of a column are admitted, the residual vanishes. In general, for the declared target, the residual measures the variation of the target that the admitted source coordinates do not capture; a source that discards some coordinates can still be adequate for a target that ignores them, as the orientation-normalized readout of Theorem 8.3 shows. Most of the adequacy collapses in this paper are elementary for exactly this reason. Their value is not depth but bookkeeping: they say which coordinates must be present before a datum can serve as a source for a given readout.

The “does not determine” results complement the collapses. Some are formal (Theorems 2.4, 9.3, 10.1 and 11.1), and their content is the precise identification of what is forgotten. One is arithmetic: the curves 571a1 and 2045b1 (Example 5.5) have the same rank, the same Tamagawa numbers, the same torsion and the same 2-torsion dimension of Sha, but 2-primary parts of Sha of orders 4 and 16. The dimension of $\text{Sha}[2]$ is the datum most directly computed by 2-descent, and the example shows concretely that it is not the datum the BSD formula needs.

Three normalization points recur and deserve emphasis. First, a Cassels–Tate pairing with values in $\mathbb{Q}/\mathbb{Z}$ does not give a canonical numerical Pfaffian. The invariant finite factor is the order of the paired group; an alternating integral presentation A gives $\#S_p = |\det A| = \text{Pf}(A)^2$, which determines $|\text{Pf}(A)|$ but not its sign (Section 5). Second, on the finite local condition the relevant map is the logarithm; the dual exponential vanishes there (Section 6). Third, the orientation enters the determinant assembly as an independent input; the assembled value must not be reused as a factor (Sections 7 and 8). Each of these is a small point, but each changes a formula.

At the two ends of the apparatus sit two calibrations. At the source end, the exterior algebra vsr_r records the shape that a rank- r source must have, and the rank-two discussion shows that a scalar leading-term identity does not by itself provide such a source. At the target end, the normalization $\kappa_r = \# \text{Sha} / \# E(\mathbb{Q})_{\text{tors}}^2$ is exactly what makes the cascade form of the leading term coincide with the standard one.

The language of typed columns, readouts and residuals comes from the Six Birds framework [36, 34, 35], but nothing in the paper depends on that framework beyond the definitions given here. The apparatus is used by the companion paper [33], which treats the scalar BSD identity as a closure problem under explicit hypotheses.

16 Conclusion

We separated the data behind the BSD formula into five columns and tested each column with a finite Schur-complement residual. The full native data of each column are adequate for its readout,

and the five columns together are adequate for the orientation-normalized Bloch–Kato scalar, which forgets the orientation coordinate. Coarser readouts lose information in precisely identified ways, and for the 2-torsion dimension of Sha this loss occurs for actual curves. The comparisons from the native columns to the Bloch–Kato line remain external inputs. The resulting picture is a checklist: for each factor of the BSD formula, which source data must be kept, and which normalizations must be fixed, before the factor can be compared across theories.

A Lean formalization and supporting data

What is formalized

The Lean 4 development accompanying this paper imports no external mathematics library. Its theorems depend only on Lean’s standard axioms (propositional extensionality, quotient soundness and classical choice, together with their standard consequences); there are no project-specific axioms, and no declaration is left unproved. The formalized content is finite and algebraic: coordinate calculations with explicit matrices, finite groups with explicit pairings, and records with tags. None of the arithmetic geometry used in the paper is formalized; where a statement depends on it, the footnote and the body say so.

Each of the 41 labelled definitions, theorems, obligations, remarks, warnings and nonclaims falls into one of five classes, used in the footnotes and in Table 3. (The example in Section 5 is not one of these statements; its inputs are described in the text.)

- *Lean proves* (7 theorems): the Lean statement has the same mathematical content as the theorem in the body.
- *Model version* (11 theorems): Lean proves a narrower statement, for instance on a model of the objects (integer coordinates, records with tags, a rank-two case), and the footnote says what is narrower. The general statement is proved in the body.
- *Conditional schema* (1 theorem, Theorem 12.5): the conclusion is obtained by projecting a supplied hypothesis. Lean checks the logical bookkeeping, not the hypothesis.
- *Lean definition* (11 definitions): the definition has a Lean counterpart. Most of these are interfaces that take the relevant arithmetic operations as inputs rather than constructing them.
- *Not formalized* (11 statements): the five transport obligations, four remarks, one warning and one nonclaim.

These classes follow the coverage records in the repository, which compare each Lean statement with the corresponding statement of the paper.

Supporting computations

Some statements use computations outside Lean. The 2-descents, modular symbols and local data for 571a1, 1309a1 and 2045b1 (Example 5.5) were computed with PARI/GP [32]. The heights of 389a1 (Section 9) are the output of the PARI/GP command

```
ellheightmatrix(ellinit([0,1,1,-2,0]), [[0,0],[1,0]]).
```

The 76-row table of Remark 11.3 was computed by a separate program that counts points of each reduction twice, by exhaustive enumeration and by counting roots of quadratics. The 2-primary computation for 2045b1 also uses the published eight-descent of Miller [21, proof of Theorem 7.1], which we did not rerun.

Data and code availability

The Lean sources, the statement-by-statement coverage records, the numerical support table, and scripts that check the descent, modular-symbol and local-data computations and recompute the support table are distributed with the archived release of this paper cited on the first page.

Coverage and declarations

Table 3. Lean coverage of the labelled statements in the inventory, in document order (41 rows). Labels omit the apparatus: infix.

Label	Kind	Lean coverage	Section
def:bk-component-map	definition	Lean definition	Section 2
def:bridge-defect-equation	definition	Lean definition	Section 2
thm:component-killing	theorem	Lean proves	Section 2
thm:finite-scalar-public-shadow	theorem	model version	Section 2
nonclaim:decomposition-scope	nonclaim	not formalized	Section 2
def:height-regulator-map	definition	Lean definition	Section 3
thm:height-schur-collapse	theorem	model version	Section 3
obl:rho-ht-reg-transport	obligation	not formalized	Section 3
warn:571a1-carrier	warning	not formalized	Section 3
def:analytic-period-map	definition	Lean definition	Section 4
thm:analytic-period-schur-collapse	theorem	Lean proves	Section 4
obl:rho-an-period-transport	obligation	not formalized	Section 4
def:finite-source-map	definition	Lean definition	Section 5
thm:finite-source-schur-collapse	theorem	Lean proves	Section 5
thm:scalar-finite-shadow	theorem	Lean proves	Section 5
thm:dim-sha-shadow	theorem	model version	Section 5
obl:rho-finite-transport	obligation	not formalized	Section 5
def:p-adic-map	definition	Lean definition	Section 6
thm:ordinary-schur-collapse	theorem	model version	Section 6
thm:signed-schur-collapse	theorem	model version	Section 6
obl:rho-p-transport	obligation	not formalized	Section 6
def:det-assembly-map	definition	Lean definition	Section 7
thm:cascade-completion-signature	theorem	Lean proves	Section 7
obl:rho-det-assembly	obligation	not formalized	Section 7
rmk:five-column-literature-gap	remark	not formalized	Section 8
def:five-column-tensor	definition	Lean definition	Section 8
thm:five-col-bk-comparison	theorem	Lean proves	Section 8
def:rank-two-kummer-source	definition	Lean definition	Section 9
thm:rank-two-height-schur-collapse	theorem	model version	Section 9
thm:gram-matrix-shadow	theorem	model version	Section 9
thm:finite-window-no-go	theorem	model version	Section 10
thm:finite-prime-cover-no-go	theorem	Lean proves	Section 11
def:support-prime-truncation-residual	definition	Lean definition	Section 11
rmk:support-prime-atlas	remark	not formalized	Section 11

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Label	Kind	Lean coverage	Section
def:vsrca-algebra	definition	Lean definition	Section 12
thm:vsrca-stage-i-consistency	theorem	model version	Section 12
rmk:vsrca-no-licensed-descent	remark	not formalized	Section 12
thm:vsrca-stage-ii-gz-partial	theorem	model version	Section 12
thm:vsrca-stage-iii-translation	theorem	conditional schema	Section 12
rmk:vsrca-rank-r-stronger	remark	not formalized	Section 12
thm:kappa-r-normalization-equivalence	theorem	model version	Section 13

Table 4. Lean declarations for the 30 formalized statements. Labels omit the `apparatus: infix;` declarations omit the `SixBirdsBSD.Apparatus. namespace.`

Label	Lean declaration
def:bk-component-map	<code>Decomposition.bkComponentMap</code>
def:bridge-defect-equation	<code>Decomposition.bridgeDefectEquation</code>
thm:component-killing	<code>Decomposition.componentKilling</code>
thm:finite-scalar-public-shadow	<code>Decomposition.finiteScalarPublicShadow</code>
def:height-regulator-map	<code>HeightRegulator.heightRegulatorMap</code>
thm:height-schur-collapse	<code>HeightRegulator.heightSchurCollapse</code>
def:analytic-period-map	<code>AnalyticPeriod.analyticPeriodMap</code>
thm:analytic-period-schur-collapse	<code>AnalyticPeriod.analyticPeriodSchurCollapse</code>
def:finite-source-map	<code>FiniteSource.finiteSourceMap</code>
thm:finite-source-schur-collapse	<code>FiniteSource.finiteSourceSchurCollapse</code>
thm:scalar-finite-shadow	<code>FiniteSource.scalarFiniteShadow</code>
thm:dim-sha-shadow	<code>FiniteSource.dimShaShadow</code>
def:p-adic-map	<code>PAdic.pAdicMap</code>
thm:ordinary-schur-collapse	<code>PAdic.ordinarySchurCollapse</code>
thm:signed-schur-collapse	<code>PAdic.signedSchurCollapse</code>
def:det-assembly-map	<code>DetAssembly.detAssemblyMap</code>
thm:cascade-completion-signature	<code>DetAssembly.cascadeCompletionSignature</code>
def:five-column-tensor	<code>Comparison.fiveColumnTensor</code>
thm:five-col-bk-comparison	<code>Comparison.fiveColBkComparison</code>
def:rank-two-kummer-source	<code>HigherRankNoGo.rankTwoKummerSource</code>
thm:rank-two-height-schur-collapse	<code>HigherRankNoGo.rankTwoHeightSchurCollapse</code>
thm:gram-matrix-shadow	<code>HigherRankNoGo.gramMatrixShadow</code>
thm:finite-window-no-go	<code>GlobalAuditNoGo.finiteWindowNoGo</code>
thm:finite-prime-cover-no-go	<code>SupportPrimeNoGo.finitePrimeCoverNoGo</code>
def:support-prime-truncation-residual	<code>SupportPrimeNoGo.supportPrimeTruncationResidual</code>
def:vsrca-algebra	<code>Vsrc.vsrcAlgebra</code>
thm:vsrca-stage-i-consistency	<code>Vsrc.vsrcStageIConsistency</code>
thm:vsrca-stage-ii-gz-partial	<code>Vsrc.vsrcStageIIGzPartial</code>
thm:vsrca-stage-iii-translation	<code>Vsrc.vsrcStageIIITranslation</code>
thm:kappa-r-normalization-equivalence	<code>KappaNormalization.kappaRNormalizationEquivalence</code>

B Version history

v1, 31 August 2026

First public version.

v2, 4 October 2026

Revised after a statement-by-statement audit of the mathematics and of its Lean formalization. Changes:

- The Sha factor of the finite column is the order of the paired group, read off an integral presentation; the numerical Pfaffian of a $\mathbb{Q}/\mathbb{Z}$ -valued pairing is not used. The $\dim \text{Sha}[2]$ result is proved for actual paired groups, and an arithmetic example (571a1 and 2045b1) is added.
- The finite local coordinate is the Bloch–Kato logarithm; the signed family is restricted to odd p with $a_p = 0$; the real readouts of p -adic data are stated as supplied inputs.
- The fifth factor of the five-column tensor is the orientation datum, not the assembled value, and the scalar comparison is stated for an orientation-normalized readout.
- The finite-shadow witness uses even Sha valuations; the height differential is stated with its off-diagonal factor 2, and the heights of 389a1 are recomputed for an explicit basis; the rank-one Heegner identity includes the index of the Heegner point; the rank-two calibration for 389a1 no longer assumes $\# \text{Sha} = 1$; the κ_r normalization states its nonzero hypotheses.
- Five small-prime supersingular rows of the support table are identified as outside the cited signed theory.
- Statements proved by Lean only in a model version are labelled as such; the literature remark is restated without an audit claim; the text is rewritten throughout for readability.

References

- [1] A. A. Beilinson. Higher regulators and values of L-functions. *J. Soviet Math.*, 30:2036–2070, 1985. doi:10.1007/BF02105861.
- [2] Massimo Bertolini, Henri Darmon, and Kartik Prasanna. Generalized Heegner cycles and p-adic Rankin L-series. *Duke Math. J.*, 162(6):1033–1148, 2013. doi:10.1215/00127094-2142056.
- [3] B. J. Birch and H. P. F. Swinnerton-Dyer. Notes on elliptic curves. II. *J. Reine Angew. Math.*, 218: 79–108, 1965. doi:10.1515/crll.1965.218.79.
- [4] Spencer Bloch. Algebraic cycles and higher K-theory. *Adv. Math.*, 61(3):267–304, 1986. doi:10.1016/0001-8708(86)90081-2.
- [5] Spencer Bloch and Kazuya Kato. L-functions and Tamagawa numbers of motives. In *The Grothendieck Festschrift, Vol. I*, volume 86 of *Progr. Math.*, pages 333–400. Birkhäuser, 1990. doi:10.1007/978-0-8176-4574-8_9.
- [6] Christophe Breuil, Brian Conrad, Fred Diamond, and Richard Taylor. On the modularity of elliptic curves over $\mathbf{Q}$: wild 3-adic exercises. *J. Amer. Math. Soc.*, 14(4):843–939, 2001. doi:10.1090/S0894-0347-01-00370-8.
- [7] David Burns and Matthias Flach. Tamagawa numbers for motives with (non-commutative) coefficients. *Doc. Math.*, 6:501–570, 2001. doi:10.4171/DM/113.
- [8] J. W. S. Cassels. Arithmetic on curves of genus 1. IV. Proof of the Hauptvermutung. *J. Reine Angew. Math.*, 211:95–112, 1962.
- [9] Kęstutis Česnavičius. The Manin constant in the semistable case. *Compos. Math.*, 154(9):1889–1920, 2018. doi:10.1112/S0010437X18007273.
- [10] J. E. Cremona. *Algorithms for Modular Elliptic Curves*. Cambridge Univ. Press, 2 edition, 1997.
- [11] Pierre Deligne. Le déterminant de la cohomologie. In *Current Trends in Arithmetical Algebraic Geometry*, volume 67 of *Contemp. Math.*, pages 93–177. Amer. Math. Soc., 1987.
- [12] Jean-Marc Fontaine. Exposé II: Le corps des périodes p-adiques. In Jean-Marc Fontaine, editor, *Périodes p-adiques – Séminaire de Bures, 1988*, number 223 in *Astérisque*, pages 59–101. Soc. Math. France, 1994. URL https://www.numdam.org/item/AST_1994__223__59_0/.
- [13] Jean-Marc Fontaine and Bernadette Perrin-Riou. Autour des conjectures de Bloch et Kato: cohomologie galoisienne et valeurs de fonctions L. In *Motives (Seattle, 1991)*, volume 55.1 of *Proc. Sympos. Pure Math.*, pages 599–706. Amer. Math. Soc., 1994.
- [14] Benedict H. Gross and Don B. Zagier. Heegner points and derivatives of L-series. *Invent. Math.*, 84(2): 225–320, 1986. doi:10.1007/BF01388809.
- [15] Roger A. Horn and Charles R. Johnson. *Matrix Analysis*. Cambridge Univ. Press, 2 edition, 2013.
- [16] Kazuya Kato. p-adic Hodge theory and values of zeta functions of modular forms. *Astérisque*, 295: 117–290, 2004.
- [17] Finn Faye Knudsen and David Mumford. The projectivity of the moduli space of stable curves. I. Preliminaries on “det” and “div”. *Math. Scand.*, 39(1):19–55, 1976. doi:10.7146/math.scand.a-11642.
- [18] Shin-ichi Kobayashi. Iwasawa theory for elliptic curves at supersingular primes. *Invent. Math.*, 152(1): 1–36, 2003. doi:10.1007/s00222-002-0265-4.

- [19] V. A. Kolyvagin. Euler systems. In *The Grothendieck Festschrift, Vol. II*, volume 87 of *Progr. Math.*, pages 435–483. Birkhäuser, 1990.
- [20] Barry Mazur. Rational points of abelian varieties with values in towers of number fields. *Invent. Math.*, 18:183–266, 1972. doi:10.1007/BF01389815.
- [21] Robert L. Miller. Proving the Birch and Swinnerton-Dyer conjecture for specific elliptic curves of analytic rank zero and one. arXiv:1010.2431v3, 2011. Published in *LMS J. Comput. Math.* 14 (2011), 327–350; theorem numbers refer to arXiv version 3.
- [22] J. S. Milne. *Arithmetic Duality Theorems*, volume 1 of *Perspectives in Mathematics*. Academic Press, 1986.
- [23] Jan Nekovář. Beilinson’s conjectures. In *Motives (Seattle, 1991)*, volume 55.1 of *Proc. Sympos. Pure Math.*, pages 537–570. Amer. Math. Soc., 1994.
- [24] Jan Nekovář. *Selmer Complexes*. Number 310 in *Astérisque*. Soc. Math. France, 2006.
- [25] André Néron. Quasi-fonctions et hauteurs sur les variétés abéliennes. *Ann. of Math. (2)*, 82:249–331, 1965. doi:10.2307/1970644.
- [26] Robert Pollack. On the p-adic L-function of a modular form at a supersingular prime. *Duke Math. J.*, 118(3):523–558, 2003. doi:10.1215/S0012-7094-03-11835-9.
- [27] Bjorn Poonen and Michael Stoll. The Cassels–Tate pairing on polarized abelian varieties. *Ann. of Math. (2)*, 150(3):1109–1149, 1999. doi:10.2307/121064.
- [28] Joseph H. Silverman. *Advanced Topics in the Arithmetic of Elliptic Curves*, volume 151 of *Graduate Texts in Mathematics*. Springer, 1994. doi:10.1007/978-1-4612-0851-8.
- [29] Joseph H. Silverman. *The Arithmetic of Elliptic Curves*, volume 106 of *Graduate Texts in Mathematics*. Springer, 2 edition, 2009. doi:10.1007/978-0-387-09494-6.
- [30] John T. Tate. On the conjectures of Birch and Swinnerton-Dyer and a geometric analog. In *Séminaire Bourbaki, Vol. 9*, pages Exp. No. 306, 415–440. Soc. Math. France, 1966.
- [31] The LMFDB Collaboration. The L-functions and modular forms database. <https://www.lmfdb.org>, 2024. Online database.
- [32] The PARI Group. PARI/GP version 2.15.4. Univ. Bordeaux, <https://pari.math.u-bordeaux.fr/>, 2023.
- [33] Ioannis Tsiokos. A Conditional Proof of the Strong Birch–Swinnerton-Dyer Conjecture, 2026. Preprint, version 2. doi:10.5281/zenodo.23130788.
- [34] Ioannis Tsiokos. Six Birds Foundations II: Admissibility Meta-Theory and the Exact Six Program, 2026.
- [35] Ioannis Tsiokos. Six Birds Foundations III: A Finite Audited Interaction Calculus for SBT, 2026.
- [36] Ioannis Tsiokos. Six Birds: The Needles Framework Apparatus (adequacy, Schur-complement diagnostics), 2026.
- [37] Andrew Wiles. Modular elliptic curves and Fermat’s last theorem. *Ann. of Math. (2)*, 141(3):443–551, 1995. doi:10.2307/2118559.
- [38] Fuzhen Zhang, editor. *The Schur Complement and Its Applications*, volume 4 of *Numerical Methods and Algorithms*. Springer, 2005. doi:10.1007/b105056.